GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

INTEGRATED TEMPERATURE SYSTEM

APPLICATION NOTES

GEK 65297A

GENERAL ELECTRIC

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SALEM, VA. 24153

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GEK 65297

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ITS APPLICATION NOTES

277A4936

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REV. 2	ISSUED Jan 25/79			FIRST MADE FOR		277A4936
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ITS APPLICATION NOTES

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A B C D E F G H J K L

01 03 05 07 09 11 13 15 17 19 21 23 25 27 29 31 33 35 37 39 41 43 45 47 49 51

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FORM IC 2390 12-74

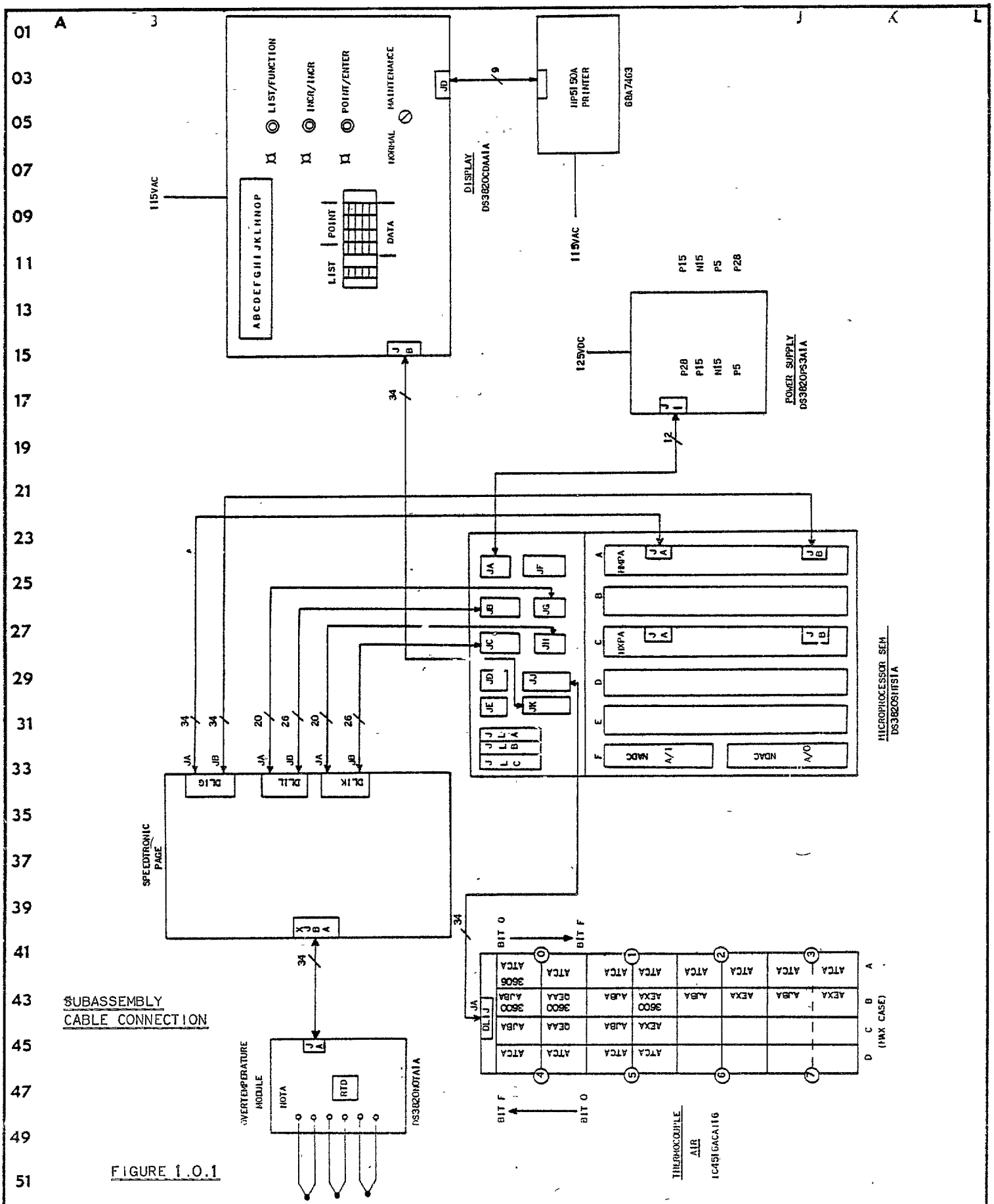


FIGURE 1.0.1

REV. 1	REV 4	REV 5	PRINTS TO	APPROVALS Azra Ehsan	GENERAL ELECTRIC
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ELEMENTARY DIAGRAM
ITS APPLICATION NOTES
2 7 7 A 4 9 3 6
CONT. ON SH. 4 SH. NO. 3

1.0.0 PURPOSE AND FUNCTION OF INTEGRATED TEMPERATURE SYSTEM (ITS)

THE INTEGRATED TEMPERATURE SYSTEM IS A MICROPROCESSOR BASED SYSTEM WHICH INTEGRATES MOST OF THE TEMPERATURE DEPENDENT FUNCTIONS OF THE GAS TURBINE. ALL OF THE TEMPERATURES ARE MEASURED BY THERMOCOUPLES WHICH ARE CONNECTED TO AN ANALOG INPUT RACK (WITH THE EXCEPTION OF THE OVERTEMPERATURE THERMOCOUPLES) THEY ARE SCANNED PERIODICALLY, CONVERTED TO DIGITAL NUMBERS, COLD JUNCTION COMPENSATED AND STORED IN THE MICROPROCESSOR MEMORY. SIMILARLY SELECTED HIGH LEVEL SIGNALS (EG. VCE, NHP, OTA, OTB, OTC) ARE ALSO CONNECTED TO THE ANALOG INPUT RACK, PERIODICALLY SCANNED, CONVERTED TO DIGITAL NUMBER AND STORED IN THE MICROPROCESSOR MEMORY. THE CONTROL AND PROTECTION FUNCTIONS ARE ALL IMPLEMENTED BY PROGRAMS IN THE PROGRAMMABLE READ ONLY MEMORY.

THE FOLLOWING BASIC FUNCTIONS ARE INCLUDED:

- THERMOCOUPLE CONDITIONING
- COMBUSTION MONITOR
- WHEELSPACE MONITOR
- OVERTEMPERATURE PROTECTION
- TEMPERATURE (FUEL) CONTROL
- DISPLAY OF SELECTED POINTS IN ENGINEERING UNITS
- DISPLAY OF PRESELECTED POINTS IN VARIOUS LISTS FORMAT.

THE FOLLOWING OPTIONAL FUNCTIONS MAY BE PROVIDED:

- PRINTER OUTPUT OF DISPLAY SELECTED POINTS
- NOZZLE CONTROL
- WATER INJECTION CONTROL
- INLET GUIDE VANE CONTROL FOR SURGE
- INLET GUIDE VANE CONTROL FOR TEMPERATURE
- INLET AIR HEATING CONTROL
- LOAD TUNNEL OVERTEMPERATURE PROTECTION
- STEAM INJECTION
- LUBE OIL TEMPERATURE LIMIT MONITOR

A PROGRAMMABLE IMPLEMENTATION OF THE TEMPERATURE SYSTEM GIVES THE FLEXIBILITY FOR CONTINUING CHANGE IN REQUISITION REQUIREMENTS. A CHANGE IN CONSTANTS OR ADDITION OF OPTIONAL FUNCTIONS IN THE SYSTEM REQUIRES A CHANGE IN THE PROM (MEMORY WHERE THE SYSTEMS PROGRAM IS STORED).

HARDWARE REQUIRED FOR THE ITS HAS BEEN DESIGNED TO BE MODULAR BY FUNCTION. THE FOLLOWING ARE THE STANDARD MODULES IN THE ITS. REFER TO FIG. 1.0.1.

(1) A SMALL ELECTRONICS MODULE (SEM), CONTAINING THE MICROPROCESSOR AND EXPANDER BOARDS, A/D AND D/A BOARDS. DS3820SMT

(2) DISPLAY MODULE CONSISTING OF OPERATOR INTERFACE SWITCHES, DISPLAY AN INDICATOR LIGHTS. DS3820CDA

(3) ANALOG INPUT RACK (AIR) CONSISTING OF TERMINATION POINTS FOR SYSTEMS THERMOCOUPLES AND THERMOCOUPLE SIGNAL CONDITIONING CARDS.

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(4) OVERTEMPERATURE TRIP MODULE. DS382ONOTA

(5) POWER SUPPLY MODULE DS382CPS3A

(6) PRINTER (OPTIONAL)

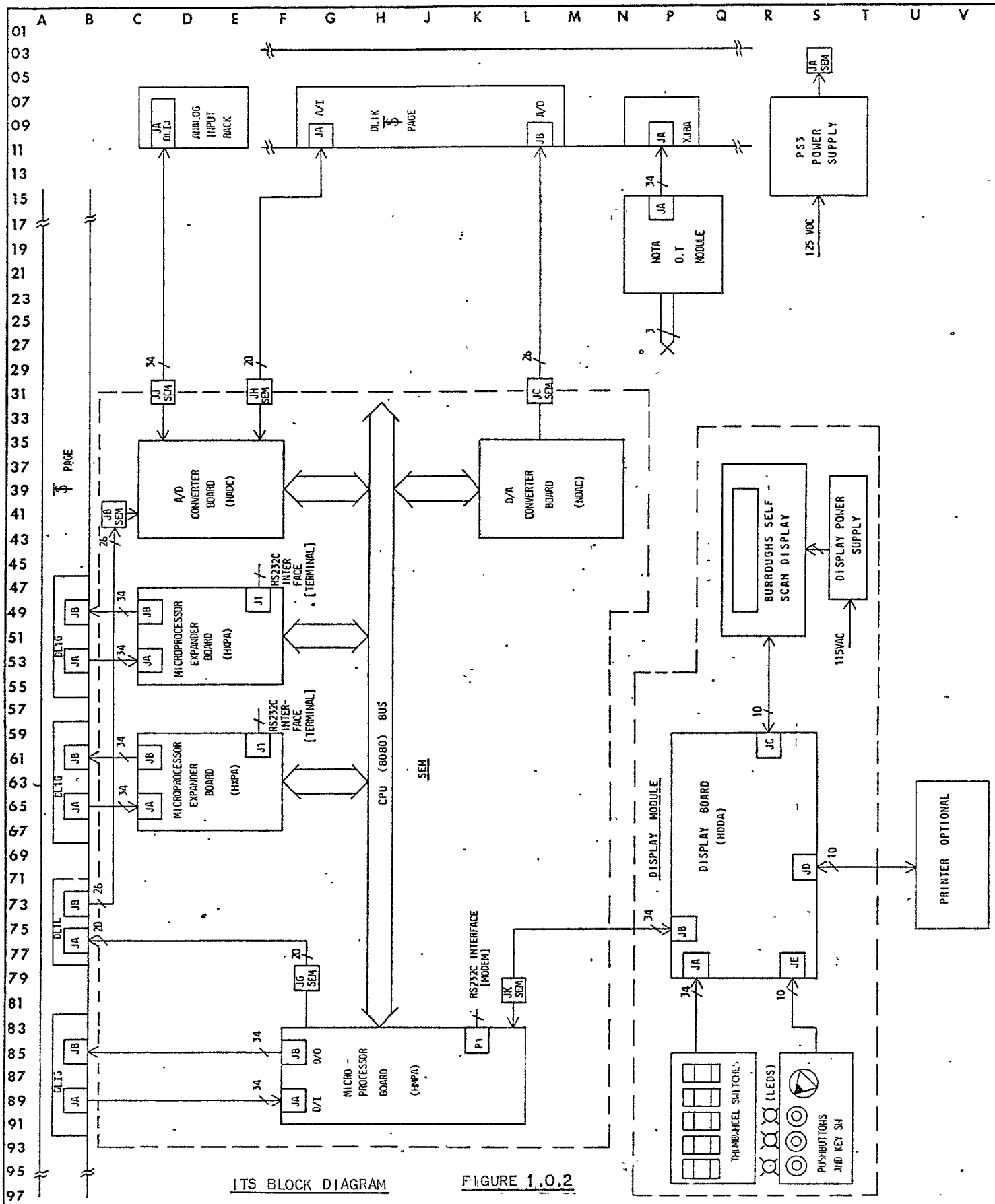
THE HARDWARD MODULES FOR ITS ARE MOUNTED IN THE TURBINE CONTROL PANEL. RIBBON CABLES ARE USED FOR INTERCONNECTION BETWEEN MODULES AND THE SPEEDTRONIC HARDWARE. REFER TO FIGURE 1.0.2.

1.0.1 START-UP

A REAL TIME AND DATE CLOCK IS PART OF ITS.
UPON POWER TURN ON OR RETURN OF THE KEY SWITCH FROM MAINTENANCE POSITION OPERATOR WILL ENTER TIME AND DATE ACCORDING TO THE PROCEDURE IN LIST C (SECTION 2.1.4).

OPERATOR WILL MAKE ADJUSTMENT IN SECONDS TO CLOCK RATE AFTER OBSERVING A LEAD OR LAG IN THE CLOCK ACCORDING TO THE PROCEDURE IN LIST C (SECTION 2.1.4).

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ITS BLOCK DIAGRAM

FIGURE 1.0.2

REV 1	REV 4	REV. 5	PRINTS TO	ENGINEER Azra Ehsan	GENERAL ELECTRIC DRIVE SYSTEMS DEPARTMENT SALEM, VA. U.S.A.
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REV 3	MADE BY AZRA EHSAN				

ELEMENTARY DIAGRAM

ITS APPLICATION NOTES

277A4936 G

CONT. ON SH 7 SH. NO 6

2.0.0 SMALL ELECTRONICS MODULE (SEM) DS3820SMT

THIS MODULE ACCOMODATES THE FOLLOWING CARDS (FIGURE 2.0.2)

-DS3800 HMPA	MICROPROCESSOR CARD
-DS3800 HXPA	MICROPROCESSOR EXPANDER CARD
-DS3800 NADC	ANALOG TO DIGITAL CARD
-DS3800 NDAC	DIGITAL TO ANALOG

ALL THE DATA PROCESSING IS HANDLED BY THIS MODULE. SIGNALS FROM THE S$_T$ PANEL AND THE THERMOCOUPLE ANALOG INPUT RACK ARE FED TO THE SEM. FROM THE DATA PASSED TO THE SEM, PROTECTIVE AND CONTROL SIGNALS ARE GENERATED WITHIN THE SEM AND SENT TO THE S$_T$ - PANEL.

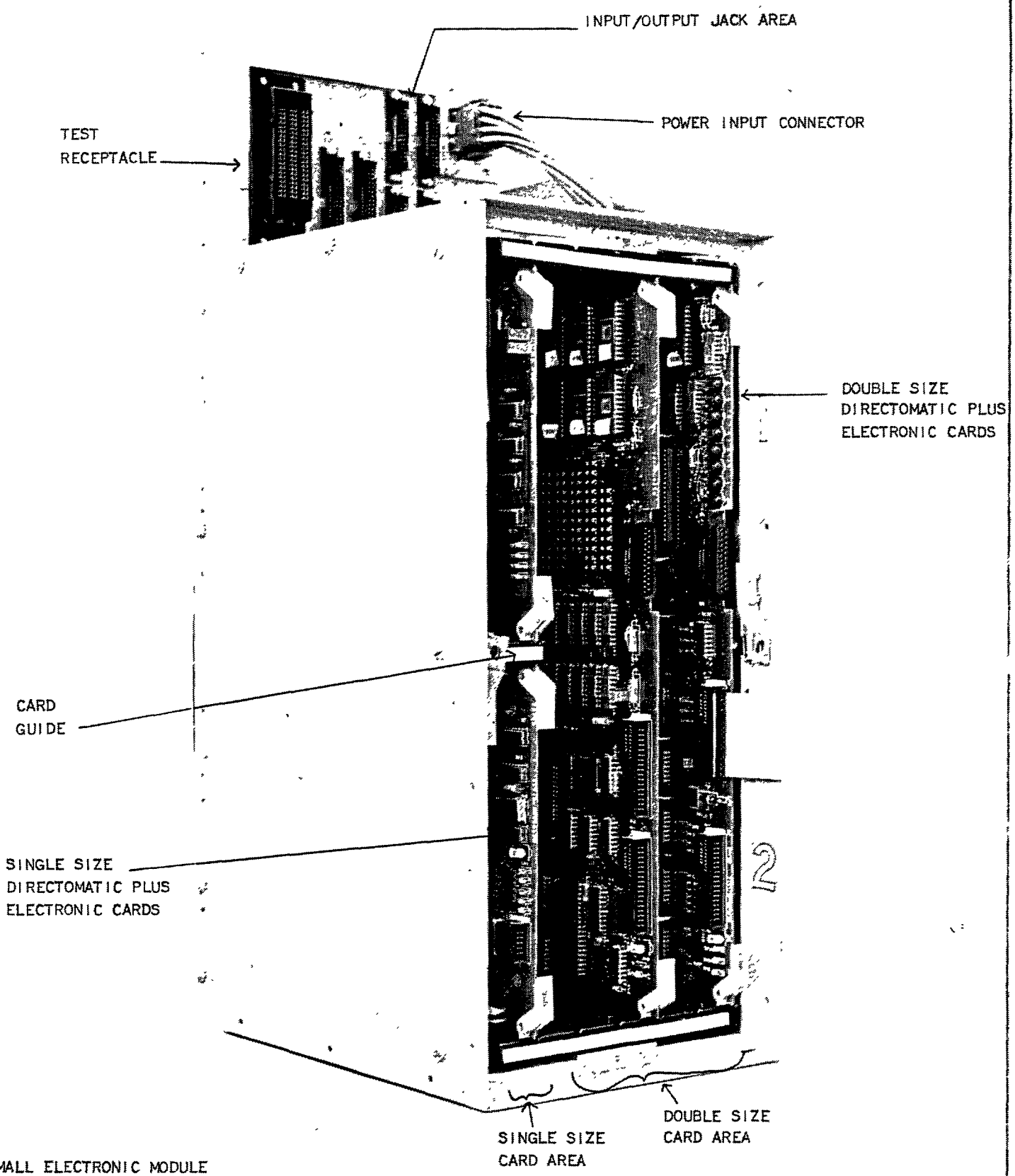
THE ITS SOFTWARE IS STORED IN 12 PROGRAMMABLE READ ONLY MEMORY CHIPS. A CHANGE IN THE CONSTANTS OR SYSTEMS OPTION MAY REQUIRE RE-PROGRAMMING OF THESE MEMORY CHIPS.

POWER TO THE SEM IS FED FROM THE PS3 POWER SUPPLY MODULE. CONNECTIONS ARE MADE TO THE POWER SUPPLY BY A CONNECTOR CABLE.

CONNECTIONS TO THE S$_T$ PANEL ARE MADE THROUGH RIBBON CABLE CONNECTORS. INTERFACE BOARDS DS3600 DLIG AND DS3600-DLIL ARE USED TO CLAMP THE INPUT AND OUTPUT SIGNALS TO +5 VDC.

A CONNECTOR FOR A DIAGNOSTIC TEST FIXTURE IS LOCATED ON THE SEM. SEE GEK 65425 FOR DETAILS ON USE OF THE DIAGNOSTIC TEST FIXTURE.

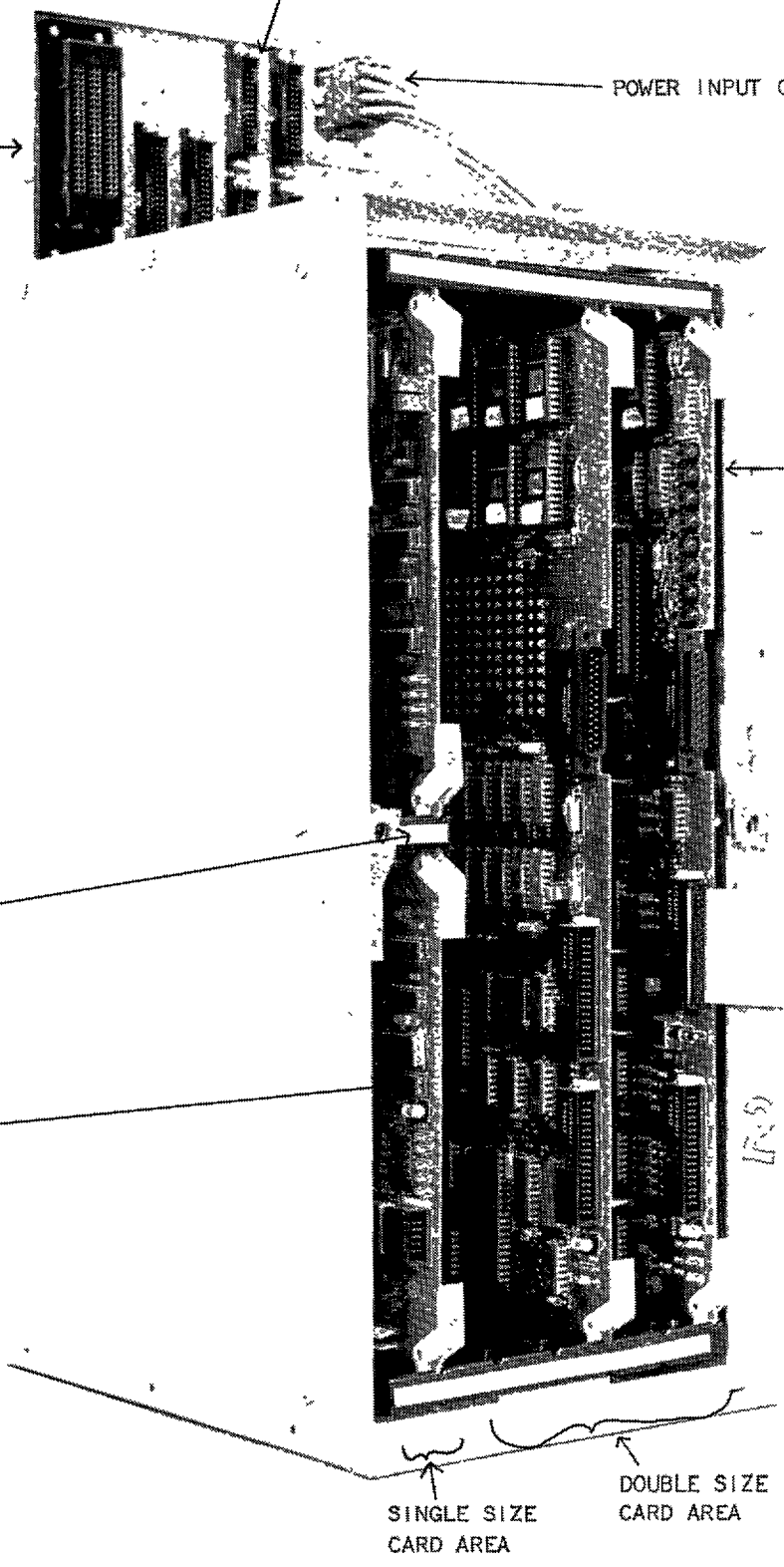
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SMALL ELECTRONIC MODULE

FIGURE 2.0.1 12-SLOT CARD RACK WITH DOUBLE-SIZE ELECTRONICS CARDS INSTALLED

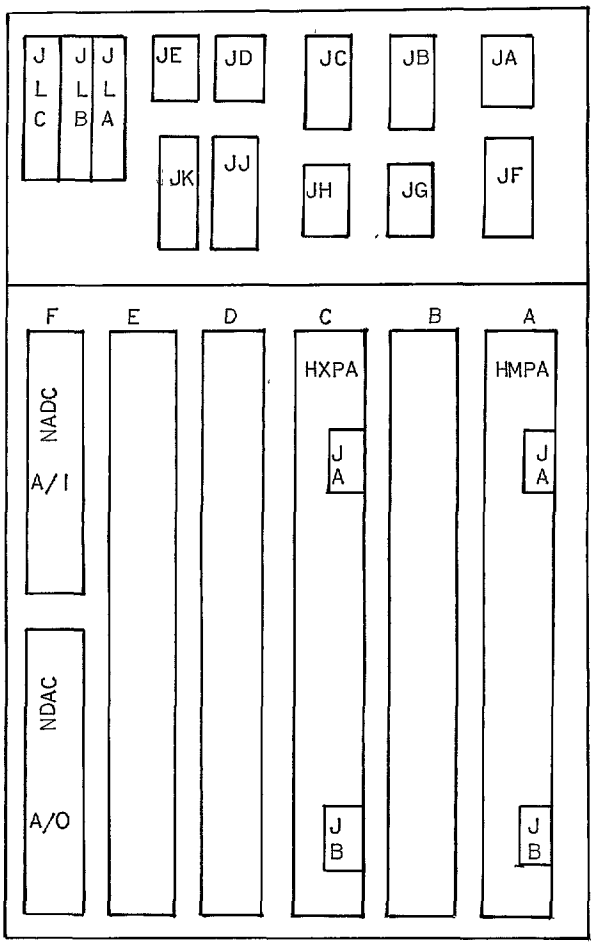
REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS Azra Ehsan	GENERAL ELECTRIC	ELEMENTARY DIAGRAM ITS APPLICATIONS NOTES
REV. 2	ISSUED 11-1-78			FIRST MADE FOR		277A4936
REV. 3	MADE BY AZRA EHSAN			I.C NO		CONT. ON SH 9 SH NO 8



SMALL ELECTRONIC MODULE

FIGURE 2.0.1 12-SLOT CARD RACK WITH DOUBLE-SIZE ELECTRONICS CARDS INSTALLED

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS Azra Ehsan	GENERAL ELECTRIC	ELEMENTARY DIAGRAM ITS APPLICATIONS NOTES
REV. 2	ISSUED 11-1-78			FIRST MADE FOR		277A4936
REV. 3	MADE BY AZRA EHSAN			I C NO		CONT. ON SH. 9 SH NO 8



MICROPROCESSOR SEM CARDS AND CONNECTORS LAYOUT

FIGURE 2.0.2

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REV. 3	MADE BY AZRA EHSAN			I C. NO.		CONT. ON SH 10 SH NO. 9

2.0.1 DS3800 NADC ANALOG INPUT CARD

THE ANALOG VOLTAGES OF THE THERMOCOUPLES ARE CONVERTED TO DIGITAL DATA ON THE NADC CARD.

THE CPU ADDRESSES THE NADC CARD ON THE ADDRESS BUS (THE SYSTEM HAS CAPABILITY FOR FOUR NADC CARDS). THE ADDRESS IS DECODED TO SELECT ONE OF THE TWO INPUT/OUTPUT (8255) CHIPS ON THE NADC CARD. THE DECODED ADDRESS ALSO SELECTS ONE OF THE THREE PORTS A, B AND C OF THE I/O CHIP.

THE I/O PORTS ARE INTERFACED WITH THE CPU BI-DIRECTIONAL DATA BUS.

I/O #1:	PORT A	THERMOCOUPLE DATA (LOWER BYTE)
	PORT B	THERMOCOUPLE DATA (UPPER BYTE)
	PORT C	ADDRESS THERMOCOUPLE POINTS ON THE ANALOG INPUT RACK
I/O #2:	PORT A	CONTROL FUNCTION DATA FROM THE CPU
	PORT B	NOT USED
	PORT C	(UPPER) OUTPUTS (NOT USED)
	PORT D	(LOWER) INPUTS A/D STATUS INFORMATION

THE CPU SELECTS A SINGLE POINT ON THE ANALOG INPUT RACK AND THIS VOLTAGE IS INPUT TO THE NADC CARD. THE A/D CONVERTER STARTS THE CONVERSION UPON RECEIVING THE CONTROL SIGNAL.

CPU CONTROL BUS SIGNALS AND CLOCK (Ø2) ARE GATED ON THE NADC TO GENERATE CONTROL AND TIMING SIGNALS.

ANALOG INPUT RANGES:		± 10 VOLTS (FULL SCALE)
OUTPUT	:	12 BIT LOGIC "0" LEVEL = .25 ± .2 VDC LOGIC "1" LEVEL = 3.9 ± 1.2 VDC
SCALE FACTOR	:	+ 10 VDC = 2048 COUNTS 0V = 0 COUNTS

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REV. 3	MADE BY AZRA EHSAN			I C NO.		CONT. ON SH. 11 SH. NO. 10

2.0.2 DS3800 NDAC ANALOG OUTPUT CARD

DIGITAL VALUES OF THE INPUT DATA ARE CONVERTED TO ANALOG VOLTAGES ON THE NDAC CARD.

THE CPU ADDRESSES THE NDAC CARD ON THE ADDRESS BUS. THE CARD ADDRESS IS DETERMINED BY APPROPRIATE IDASL3-7 LINES LEFT OPEN FOR CORRESPONDING OBAPR3-7 LINES (OF THE ADDRESS BUS) REQUIRED TRUE "0". LINES OBAPRO-2 SELECT ONE OF THE FOUR OUTPUTS OF THE CARD.

A PARITY CHECK CIRCUIT IS DESIGNED ON THE BOARD. GOOD PARITY IS INDICATED BY THE YELLOW LED BEING LIT. IN THE ITS APPLICATION THE PARITY CIRCUIT HAS BEEN DISABLED AND THE YELLOW LED STAYS ON.

TEST POINTS PER OUTPUT ARE LOCATED AT THE FRONT OF THE CARD.

ANALOG OUTPUT RANGES: +9.9951V TO -10.00V

2.0.3 DS3800 HMPA - MICROPROCESSOR CARD

MAJOR COMPONENTS ON THIS CARD ARE:

1. 8 BIT PARALLEL DATA MICROPROCESSOR (INTEL 8080).
2. 4096 BYTES OF ULTRA VIOLET ERASABLE MEMORY PROM. (INTEL 2708)
3. 1024 BYTES OF READ/WRITE MEMORY RAM. (INTEL 2111)
4. SYSTEMS TIMING CIRCUITRY AND SYSTEMS CONTROL. (INTEL 8238)
5. I/O CHIPS (INTEL 8255).
6. EIGHT LEVEL PRIORITY INTERRUPT. (INTEL 8259)
7. PARITY GENERATOR
8. USART (INTEL 8251)
9. INTERFACE CIRCUITRY BETWEEN THE MICROPROCESSOR AND THE MEMORY AND AND I/O CHIPS.

THE SYSTEMS MAIN PROGRAM IS STORED IN THE 4K ULTRA VIOLET ERASABLE MEMORY (PROM). THE 1K RAM MEMORY IS USED FOR TEMPORARY STORAGE OF DATA FROM THE MICROPROCESSOR OR ANY OF THE I/O DEVICES. THE TURBINE THERMOCOUPLE READINGS ARE STORED IN THIS MEMORY.

THE CLOCK GENERATOR/DRIVER CIRCUIT CONSISTS OF A CRYSTAL OSCILLATOR, A FOUR BIT COUNTER AND GATING CIRCUITRY. TWO OUT OF PHASE CLOCK PULSES ARE GENERATED FOR THE MICROPROCESSOR TIMING REFERENCE.

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DATA IS TRANSMITTED BETWEEN THE MICROPROCESSOR MEMORY AND THE I/O DEVICES ON A BI-DIRECTIONAL 8-BIT TIME SHARE BUS. INFORMATION IS GATED ON OR OFF THE DATA BUS BY SELECTING OR DESELECTING (TRI-STATING) MEMORY AND I/O DEVICES. (TRI-STATE IS THE ABILITY TO FORCE THE OUTPUT DRIVERS INTO A HIGH IMPEDANCE MODE AS IF IT WERE REMOVED FROM THE CIRCUIT).

A 16 BIT UNIDIRECTIONAL ADDRESS BUS IS USED FOR ADDRESSING MEMORY OR I/O PORTS.

A SET OF UNIDIRECTIONAL CONTROL LINES EMANATE FROM THE MICROPROCESSOR. THESE INDICATE THE TYPE OF ACTIVITY IN PROCESS LIKE MEMORY READ OR MEMORY WRITE ETC.

TWO I/O CHIPS ARE MOUNTED ON THE HMPA CARD TO INTERFACE WITH OPERATOR DISPLAY PANEL INPUTS (THUMBWHEEL SWITCHES, RESET SWITCH) AND OUTPUTS (STATUS LEDS, NUMERIC DISPLAY). DISPLAY DATA IS TRANSFERRED TO OR FROM THE PROCESSOR AND THE I/O PORTS OF THE TWO I/O CHIPS.

I/O #1:	PORT A	OUTPUT TO $\frac{S}{T}$ PANEL.
	PORT B	DATA OUTPUT TO DISPLAY
	PORT C	(LOWER) CONTROL OUTPUT TO DISPLAY
	PORT C	(UPPER) DATA INPUT FROM DISPLAY
I/O #2:	PORT A	INPUTS FROM $\frac{S}{T}$ TO SEM.
	PORT B	OUTPUTS TO $\frac{S}{T}$
	PORT C	(LOWER) OUTPUTS (NOT USED)
	PORT C	(UPPER) INPUTS (NOT USED)

DATA COMMUNICATION IS HANDLED BY THE USART (INTEL #8251) ON THE HMPA CARD. THE USART ACCEPTS PARALLEL DATA FROM THE CPU AND TRANSMITS SERIAL STREAM OF DATA. IT ACCEPTS SERIAL DATA INPUT AND SENDS PARALLEL DATA TO THE PROCESSOR.

AN EIGHT LEVEL PRIORITY INTERRUPT CHIP (INTEL 8259) IS USED TO DETERMINE THE HIGHEST PRIORITY INTERRUPT REQUESTED AND ISSUE AN INTERRUPT REQUEST TO THE PROCESSOR. MAXIMUM UTILIZATION OF THE MICROPROCESSOR CAPACITY IS ACHIEVED BY THE INTERRUPT CHIP WHEN A TERMINET® TERMINAL OR AN RS232 COMPATIBLE TELETYPE® TERMINAL IS BEING USED. THE MICROPROCESSOR CAN RETURN TO SOME OTHER TASK AND WAIT FOR AN INTERRUPT FROM THE TERMINET OR TELETYPE WHEN ITS READY.

A PARITY GENERATOR CIRCUIT IS ALSO LOCATED ON THE HMPA CARD. GOOD PARITY IS INDICATED BY THE YELLOW LED BEING LIT. THIS CIRCUIT HAS BEEN DISABLED IN THE ITS APPLICATION AND THE YELLOW LED STAYS LIT.

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THE INTEL 8080 CHIP REQUIRES +12VDC, +5VDC AND -5VDC. THE +12VDC IS TAPPED OFF THE +15VDC POWER SUPPLY BUS. THE -5VDC IS TAPPED OFF THE -15VDC POWER SUPPLY BUS. +5VDC IS AVAILABLE ON THE SEM BUS. REFER TO INTEL MANUALS FOR HARDWARE DETAILS.

2.0.4 DS3800 HXPA - MICROPROCESSOR EXPANDER CARD

THE HXPA (EXPANDER) IS A DOUBLE SIZE PRINTED WIRE BOARD DESIGNED AS AN EXPANDER FOR THE HMPA.

MAJOR COMPONENTS ON THIS CARD ARE

1. 8K BYTES OF ROM
2. 2K BYTES OF RAM
3. 8 PRIORITY INTERRUPTS WHICH MAY BE SLAVED TO THE MASTER CHIPS ON THE HMPA
4. TWO PROGRAMMABLE TIMERS.
5. ONE USART WITH PROGRAMMABLE BAUD RATE CLOCK
6. ONE 8255 I/O CHIP
   PORT A - 8 BUFFERED INPUTS
   PORT B - 8 BUFFERED OUTPUTS
   PORT C -
   BITS 0,1,3 OUTPUT ONLY
   BITS 4,6,7 INPUT ONLY
   BITS 2,5 INPUT OR OUTPUT
   BIT 2 NON INVERTING INPUT

THE 8K BYTES OF ROM AND 2K BYTES OF RAM ARE ADDRESSABLE FROM HMPA VIA ADDRESS BUS. OUTPUT DATA IS BUFFERED TO THE HMPA BUS LINES BY TRI-STATE DRIVERS.

STRAPPING IS PROVIDED TO ENABLE THE USE OF 3 HXPA CARDS PER HMPA.

THE I/O ADDRESSING IS DETERMINED BY CONNECTIONS TO THE HMPA OR A DECODER STRAP ON THE HXPA.

THE 8 MINOR SELECT LINES ARE WIRED IDENTICAL TO THE HMPA BOARD, AND THUS THE I/O ADDRESSES DIFFER FROM THE HMPA I/O ADDRESSES BY THE CARD SELECT SECTION OF THE ADDRESS.

THE PRIORITY INTERRUPT IS USED AS A SLAVE TO THE HMPA. STRAPPING IS REQUIRED DEPENDING ON THE NUMBER OF CARDS IN THE SYSTEM. THIS STRAPPING ENABLES THE ADDRESSED SLAVE TO SEND ITS PROGRAMMED ADDRESS FOR THE REQUESTING INPUT OUT TO THE 8080 DATA BUS (ENABLES OUTPUT TRI-STATE DRIVES).

THE TIMER IS AN INTEL 8253 CHIP. THE FIRST OUTPUT IS USED AS A BAUD RATE CLOCK FOR THE SERIAL INTERFACE. Ø2 CLOCK OF THE HMPA IS CONNECTED TO THIS DEVICE.

DATA COMMUNICATION IS HANDLED BY THE USART (INTEL #8251) ON THE HMPA CARD. THE USART ACCEPTS PARALLEL DATA FROM THE CPU AND TRANSMITS SERIAL STREAM OF DATA. IT ACCEPTS SERIAL DATA INPUT AND SENDS PARALLEL DATA TO THE PROCESSOR. AND RS232 TYPE CONNECTOR IS LOCATED ON THTS BOARD.

A YELLOW PARITY LIGHT IS ON WHEN I/O PARITY GOOD OR DISABLED.

MAINTENANCE OF THIS CARD WILL BE HANDLED BY SOFTWARE DIAGNOSTICS RUN ON THE HMPA. THERE ARE NO MAINTENANCE TEST POINTS LOCATED ON THE HXPA CARD.

POWER REQUIRED

P5	1.2A
P15	0.3A
N15	0.15A

FOR DETAILS OF OPERATION OF INTEL CHIPS REFER TO INTEL USERS MANUAL OR APPLICATION NOTES.

DETAILS OF STRAPPING FOR FUNCTIONS INDICATED MAY BE OBTAINED IN CARD ELEMENTARIES GEK# (TO BE ISSUED).

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2.0.1 CONTROL DISPLAY MODULE DS3820CDA

THE PURPOSE OF THIS MODULE IS TO PROVIDE AN OPERATOR INTERFACE TO THE INTEGRATED TEMPERATURE SYSTEM.

THE MODULE INCLUDES:

1. DISPLAY BOARD (DS3800 HDDA)
2. DISPLAY POWER SUPPLY

OPERATOR INTERFACE:

3) KEY SWITCH
4) THREE PUSH BUTTONS
5) THREE INDICATING LEDS (SOFTWARE CONTROLLED)
6) FIVE HEXIDECIMAL THUMBWHEEL SWITCHES
7) A 16 CHARACTER ALPHANUMERIC DISPLAY

THE CONTROL DISPLAY MODULE IS MOUNTED ON THE FRONT DOOR OF THE TURBINE CONTROL PANEL BY FOUR SCREWS. CONNECTIONS TO THE CONTROL DISPLAY MODULE ARE MADE TO TB1, AND JD, JB CONNECTORS ON THE DS3800HDDA OF THE MODULE.

TB1 - INPUT POWER 110VAC TO 120VAC, 60HZ
JB - 34 PIN RIBBON CABLE CONNECTOR TO THE SEM CONNECTOR JK
JD - CONNECTION TO THE PRINTER (OPTIONAL)

THE CONNECTORS JA, JE, JC OF THE HDDA CARD OF THIS MODULE ARE FOR INTERCONNECTIONS INTERNAL TO THE MODULE AND SHOULD NOT BE CONSIDERED IN CONNECTION OF THIS MODULE TO A SYSTEM. REFER TO FIGURE 2.1.2 .

REFER TO BLOCK DIAGRAM, FIGURE 2.1.3 . ALL SIGNALS AT THE I/O TERMINALS OF THE DISPLAY PANEL WILL BE ZERO VOLTS TRUE.

16 SIGNAL LINES WILL CONNECT TO THE PROCESSOR. 12 INCOMING AND 4 OUTGOING. THE 4 OUTGOING LINES ARE USED TO ENABLE THE PROCESSOR TO READ THE THUMBWHEEL SWITCHES AND PUSH BUTTONS.

4 INCOMING CONTROL LINES WILL CARRY:

1. DATA PRESENT STROBE FOR THE DISPLAY.
2. CLEAR FOR THE DISPLAY.
3. BLANK FOR THE DISPLAY.
4. DATA AVAILABLE STROBE FOR THE PRINTER.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS Azra Ehsan	GENERAL ELECTRIC	ELEMENTARY DIAGRAM ITS APPLICATION NOTES
REV. 2	ISSUED Jan 25/79			FIRST MADE FOR		277A4936
REV. 3	MADE BY AZRA EHSAN			I.C. NO.		CONT. ON SH 16 SH. NO. 15

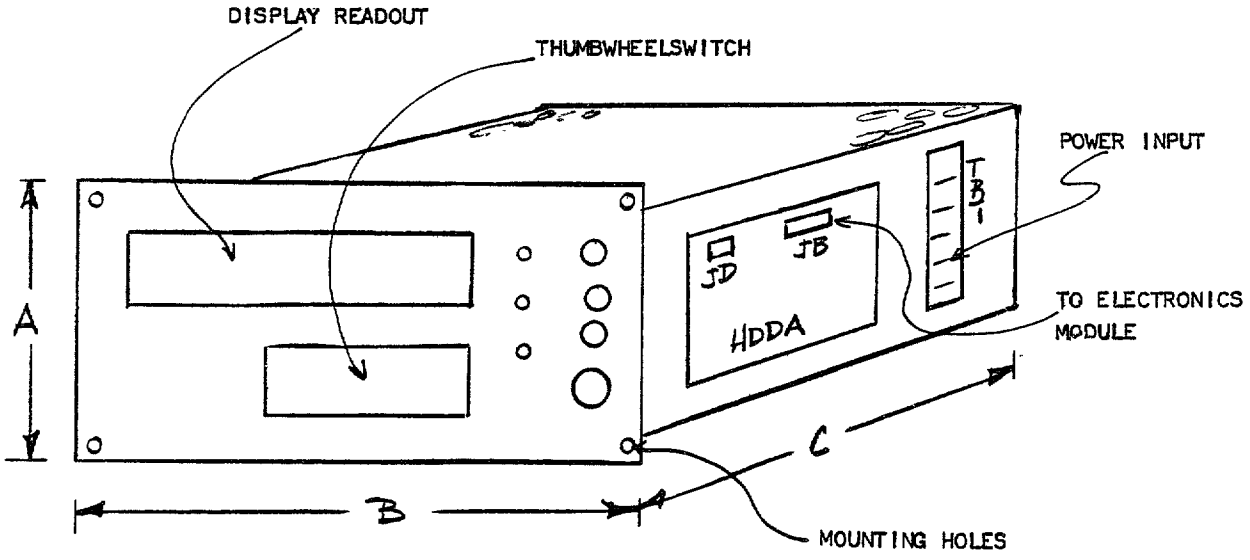
DIMENSIONS

A. 7.50"
B. 13.00"
C. 9.88"

WEIGHT ≈ 16 LBS

MOUNTING .

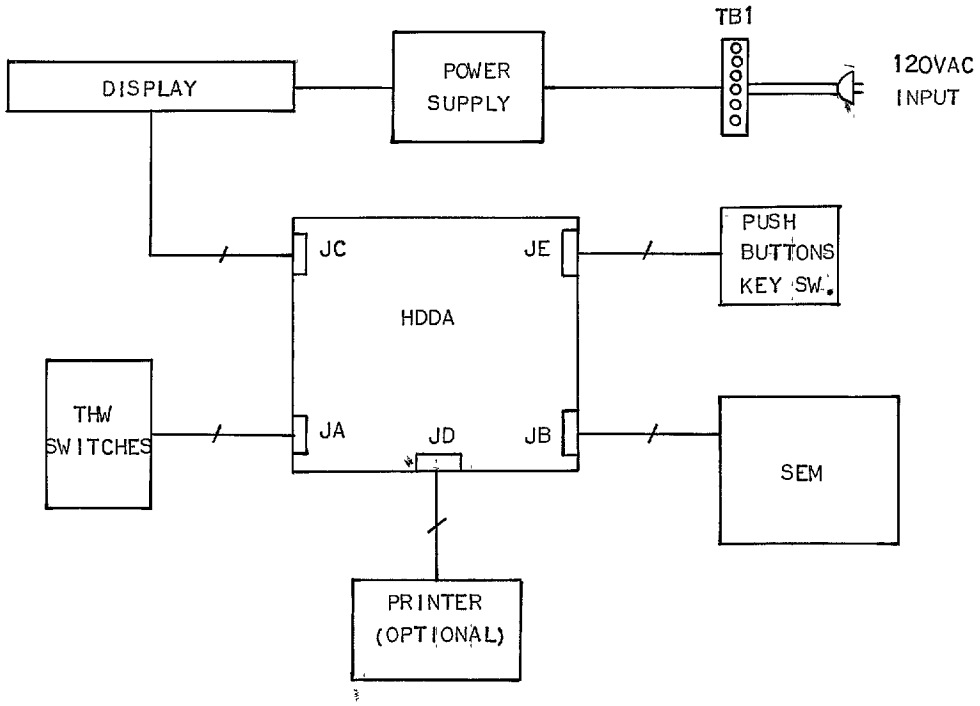
1. THE MODULE IS MOUNTED BY FOUR (4) NO.10 SCREWS THROUGH A 6.50"H X 12.50"W CUTOUT IN PANEL FRONT.



CONTROL DISPLAY MODULE ASSEMBLY

FIGURE 2.1.1

REV. 1	REV. 4	REV. 7	PRINTS TO	ENGINEER Azra Ehsan	GENERAL ELECTRIC	ITS APPLICATION NOTES
REV. 2	REV. 5	ISSUED Jan 25/79			DRIVE SYSTEM	277A4936
REV. 3	REV. 6	MADE BY AZRA EHSAN			SALEM, VA. U.S.A.	CONT. ON SH. 17 SH. NO. 16



CONTROL MODULE DISPLAY INTERCONNECT

FIGURE 2.1.2

REV. 1	REV 4	REV 5	PRINTS TO	APPROVALS Azra Ehsan	GENERAL ELECTRIC	ELEMENTARY DIAGRAM ITS APPLICATION NOTES
REV. 2	ISSUED June 25/79			FIRST MADE FOR		277A4936
REV. 3	MADE BY AZRA EHSAN			I.C. NO.		CONT. ON SH. 18 SH. NO. 17

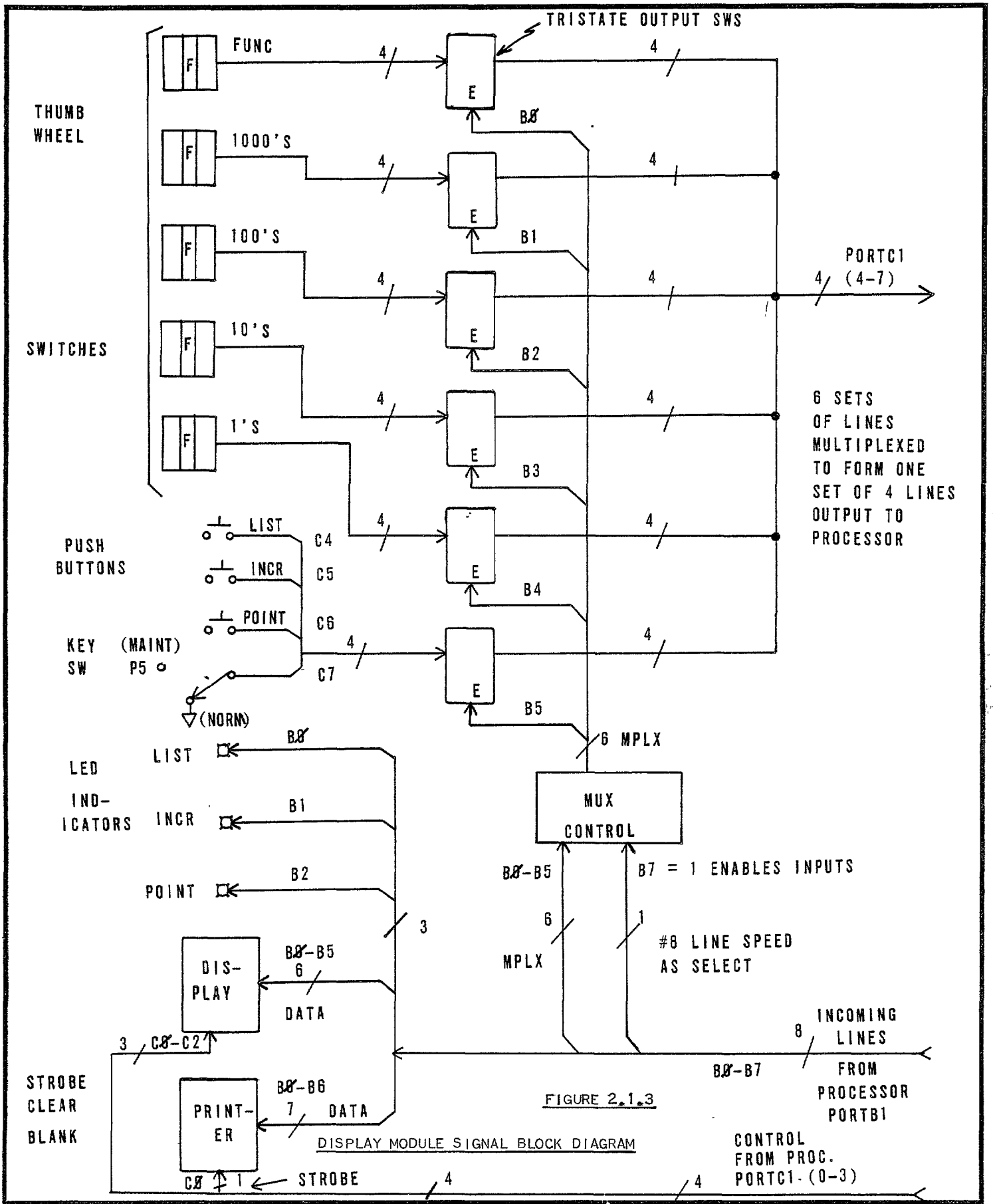


FIGURE 2.1.3

DISPLAY MODULE SIGNAL BLOCK DIAGRAM

REV. 1	REV. 4	REV. 7	PRINTS TO	ENGINEER Azra Ehsan	GENERAL ELECTRIC	ITS APPLICATION NOTES
REV. 2	REV. 5	ISSUED Jan 25/79			DRIVE SYSTEMS DEPT.	2 7 7 A 4 9 3 6
REV. 3	REV. 6	MADE BY AZRA EHSAN			SALEM, VA. U.S.A.	CONT. O.N SH. 19 SH. NO. 18

DSD 0151 (09-76)

2.1.1 DS3800 HDDA DISPLAY DRIVER CARD

THIS CARD MOSTLY CONSISTS OF TRI-STATE BUFFER INTERFACE BETWEEN THE CPU OF THE SYSTEM AND THE OPERATOR INTERFACE OF THE DISPLAY MODULE.

FOLLOWING ARE THE KEY FUNCTIONS OF THIS CARD (REFER TO SHEETS 36P, 36Q, AND 36R OF THE ITS SYSTEMS ELEMENTARY)

1. IT MULTIPLEXES SIGNALS FROM THE FIVE THUMBWHEEL SWITCHES, THE THREE PUSH BUTTONS AND THE KEY SWITCH ONTO ONE 4-BIT INPUT PORT (UPPER C-PORT I/O #1) OF THE CPU.

2. IT PROVIDES DRIVE FOR BURROUGHS SELF SCAN DISPLAY, CONSISTING OF 6 BITS OF DATA AND THREE CONTROL LINES (DATA PRESENT, CLEAR, BLANK).

3. IT PROVIDES 7 DATA LINES, ONE CONTROL LINE OUTPUT FOR THE PRINTER.

4. THREE SOFTWARE CONTROLLED LEDS INDICATING THE MODE OF OPERATION.

POWER TO THE HDDA CARD COMES FROM SEM BACK PLANE VIA RIBBON CABLE.

2.1.2 DISPLAY POWER SUPPLY

THIS IS A HIGH VOLTAGE SUPPLY USED FOR THE BURROUGH S DISPLAY. NOMINAL INPUT VOLTAGE 115VAC MAXIMUM VOLTAGE 130VAC MAXIMUM CURRENT .2 AMPS.

THIS HAS THE FOLLOWING REGULATED OUTPUTS:

P5 @ +.9 AMPS MAXIMUM
N12 @ -.175 AMPS MAXIMUM
P250 @ +.03 AMPS MAXIMUM.

8 INCOMING LINES WILL PERFORM FOUR FUNCTIONS:

A. LINES 1 TO 3 WILL LIGHT THE L.E.D. INDICATORS WHEN REQUIRED.

B. LINES 1 TO 6 ARE USED AS ADDRESSES TO MULTIPLEX THE THUMBWHEEL AND PUSHBUTTON SIGNALS ONTO THE 4 OUTGOING LINES. LINE 8 WILL BE USED AS A STROBE TO ENABLE THE MULTIPLEX CONTROL. THIS LINE MUST BE TRUE WHEN ANY INFORMATION IS BEING REQUESTED BY THE PROCESSOR.

C. LINES 1 TO 6 SEND DATA TO THE DISPLAY AS REQUIRED AND WILL PRECEED AND OVERLAP THE DATA PRESENT STROBE LINE, SENT AS A MIN. 1 MICRO SECOND PULSE.

D. LINES 1 TO 7 SEND DATA TO THE PRINTER AS REQUIRED AND WILL PRECEED AND OVERLAP THE DATA A AVAILABLE STROBE LINE TO THE PRINTER. 1/3 SEC. MUST BE ALLOWED BETWEEN LINE TRANSMISSIONS.

NOISE IS KEPT TO A MINIMUM BY INTERFACING ALL LINES INTO THE DISPLAY PANEL WITH SCHMITT TRIGGERED INVERTERS. ALL MECHANICAL SWITCHES HAVE RC NOISE FILTERS AND FEED OUT THROUGH TRISTATE OUTPUT DRIVERS HAVING SCHMITT TRIGGERED INPUTS.

POWER AND COOLING REQUIREMENTS:

1. +5V DC FOR LOGIC AND CONTROL MUST BE SUPPLIED WITH THE CONTROL CABLE FROM THE CONTROL PROCESSOR UNIT.

2. 115 OR 230V +5, -10% 48-63HZ MUST BE SUPPLIED TO OPERATE THE DISPLAY POWER SUPPLY AND THE OPTIONAL PRINTER.

3. POWER DISSIPATION
   5V - 1.5 WATTS
   110V/230V - 60HZ - 25.0 WATTS

INPUT OUTPUT SIGNAL CHARACTERISTICS

INPUTS

P5V VOLTAGE BUS
-0.4MA LOW
20 MICRO A HIGH
LOGIC "0" = 0.4V ± 0.4V
LOGIC "1" = 2.0V TO VCC

OUTPUTS

2V MIN. HIGH
0.4V MAX. LOW
24MA LOW LEVEL
-15MA HIGH LEVEL

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS Azra Ehsan	GENERAL ELECTRIC	ELEMENTARY DIAGRAM ITS APPLICATION NOTES
REV. 2	ISSUED Jan 25/79			FIRST MADE FOR		277A4936
REV. 3	MADE BY AZRA EHSAN			I.C. NO.		CONT. ON SH. 20 SH. NO 19

2.1.3 OPERATOR INTERFACE

THIS FUNCTION CONSISTS OF THE FOLLOWING:

1. ONE 16 CHARACTER ALPHA NUMERIC DISPLAY FOR OPERATOR INFORMATION.

2. TWO THUMBWHEEL SWITCHES:
   A. 16 DIGIT LIST/FUNCTION SWITCH
   B. A FOUR SECTION (16 DIGITS PER SECTION) DATA SWITCH.

3. THREE OPERATORS PUSH BUTTONS: (NORMALLY OPEN)
   A. LIST/FUNCTION
   B. INCREMENT
   C. POINT/ENTER

4. ONE KEY SWITCH MARKED "NORMAL-MAINTENANCE" .

5. THREE INDICATING LIGHTS TO ENABLE THE PROCESSOR TO INDICATE WHEN IT HAS RECIEVED ONE OF THE P.B. SIGNALS MOUNTED ON THE HDDB CARD.

6. OPTIONAL THERMAL PRINTER TO PRINT DATA SUPPLIED BY THE CENTRAL PROCESSOR WHEN TURNED ON.

PROCEDURE FOR USING ITS OPERATOR'S PANEL WITH OPTIONAL PRINTER ARE IN THE NEXT SECTION.

A PERIODIC TEST OF THE MODULES DISPLAYS SHOULD BE MADE TO CHECK FOR BURNED OUT LEDS AND DISPLAY CHARACTERS. THIS IS ACCOMPLISHED BY LOADING ITS SELF VERIFICATION DIAGNOSTICS SOFTWARE IN AN ON LINE SYSTEM ENVIRONMENT AND REFERRING TO THE ITS SELF VERIFICATION DIAGNOSTICS USERS GUIDE GEK 65317 .

2.1.3-1 BURROUGHS SSD1000-0041, ALPHA NUMERIC SELF SCAN DISPLAY WITH MEMORY

THIS UNIT IS A SINGLE ROW, SIXTEEN POSITION DISPLAY WITH ASSOCIATED DRIVE ELECTRONICS. THE DISPLAY HAS A REPERTOIRE OF SIXTY-FOUR CHARACTERS AND CONTAINS A REFRESH MEMORY. EACH CHARACTER IS DISPLAYED IN A FIVE-BY-SEVEN DOT MATRIX FORMAT WITH TWO COLUMNS OF SPACE BETWEEN CHARACTERS. EACH CHARACTER IS DEFINED BY A POSITIVE LOGIC SIX BIT CODE. THE DISPLAY IS COMPLETELY SELF CONTAINED AND PACKAGED IN A MOLDED HOUSING WITH AN INTEGRAL BEZEL.

THE DISPLAY OPERATES IN A SEQUENTIAL MODE ENTERING CHARACTERS FROM LEFT TO RIGHT. THE APPROPTIATE SIX BIT CODE MUST BE PRESENT APPROXIMATELY 33 μSEC FOR EACH ENTRY. THE DISPLAY WILL CONTINUE IN ORDER, ENTERING A TOTAL OF 16 CHARACTERS. CHARACTER NUMBER 17 WILL BE ENTERED IN THE DISPLAY SUCH THAT IT REPLACES THE FIRST CHARACTER ON THE LEFT. THE DISPLAY MAY BE CLEARED AT ANY TIME DURING THE 16 CHARACTER CYCLE BY PRESENTING THE APPROPRIATE SIGNAL AT THE CLEAR LINE FOR APPROXIMATELY 33 μSEC.

A BLANK DISABLE LINE IS PROVIDED TO BLANK THE DISPLAY WITHOUT ALTERING THE CONTENTS OF MEMORY.

2.1.3-2 PRINTER (OPTIONAL)

THE HEWLETT-PACKARD 5150A PRINTER INTERFACES THE CPU THROUGH THE HDDA CARD. THE PRINTER IS SET TO THE LISTEN ONLY MODE BY AN ADDRESSABLE SWITCH ON THE REAR OF THE PRINTER. THIS MEANS THAT THE PRINTER WILL NOT REQUIRE AN ADDRESS THROUGH THE IEE 488 INTERFACE. ALSO THE "ATN" (ATTENTION) LINE IS STRAPPED TO A "1" THIS MEANS THAT THE PRINTER WILL ALWAYS ACCEPT CHARACTERS AND NOT INTERPRET INFORMATION ON THE DATA LINES AS ADDRESS.

PRINTER TIMING IS COMPATIBLE WITH DISPLAY TIMING FOR CHARACTER BURSTS, HOWEVER, THE MAXIMUM LINE PRINTING RATE IS 3 LINES PER SECOND.

PRINTER WILL PRINT WHEN LINE ON/OFF SWITCH IS IN THE ON POSITION.

POWER FOR THE PRINTER IS 115VAC, AT 60HZ. THE PRINTER IS MOUNTED ON THE TURBINE CONTROL PANEL DOOR IN THE SPACE NORMALLY OCCUPIED BY THE THERMOCOUPLE AVERAGING MODULE. THE SPACE AVAILABLE FOR THE PRINTER AND DISPLAY MODULE IS 20 INCHES HIGH, 15 INCHES WIDE AND 15 INCHES DEEP.

THE PRINTER HAS CAPABILITY OF PRINTING FULL 64-CHARACTER UPPERCASE ASCII PRINTING SET.

FOR MORE DETAILS ON THE PRINTER REFER TO THE HEWLETT-PACKARD 5150A PRINTER MANUAL.

2.1.4 PROCEDURE FOR USING ITS OPERATORS PANEL

1. NORMAL OPERATION - THIS MODE IS SELECTED BY PUTTING THE KEY SWITCH IN NORMAL POSITION. THIS ACTION CAUSES THE ITS MODULE TO RE-INITIALIZE.

   A. LIST DISPLAY - LIST/DISPLAY MODE MAY BE ENTERED FROM POINT DISPLAY MODE BY PRESSING THE LIST PUSHBUTTON.

      1. ENTER DESIRED LIST NUMBER ON LIST THUMBWHEEL. LIST NUMBER IS HEXADECIMAL.

      2. PRESS LIST PUSHBUTTON. LIST LIGHT COMES ON. POINT AND INC. LIGHTS GO OFF IF THEY ARE ON.
      LIST DISPLAY BEGINS WITH FIRST POINT OF LIST AND PROCEEDS, DISPLAYING EACH POINT FOR TWO SECONDS (EXCEPT IN THE CASE OF LIST "E" WHERE EACH POINT WILL BE DISPLAYED FOR 1/2 SECOND IF PRINTER IS USED AND WILL BE DISPLAYED FOR 2 SECONDS IF PRINTER IS NOT USED). DATA FOR ENTIRE LIST IS CAPTURED BEFORE THE FIRST POINT IS DISPLAYED ON EACH CYCLE THROUGH THE LIST.
      DISPLAY CONTINUES UNTIL LIST THUMBWHEEL IS CHANGED (OR UNTIL KEY IS SWITCHED OR POINT FUNCTION BUTTON IS PRESSED).

      3. CHANGING LIST THUMBWHEEL CAUSES A NEW LIST TO BEGIN WITH FIRST POINT WITHIN 1/2 SECOND. NON-EXISTENT LIST NUMBERS BLANK THE DISPLAY.

      4. PRESSING LIST PUSHBUTTON AT ANY TIME CAUSES A NEW SET OF DATA TO BE CAPTURED AND THE LIST TO BEGIN AGAIN.

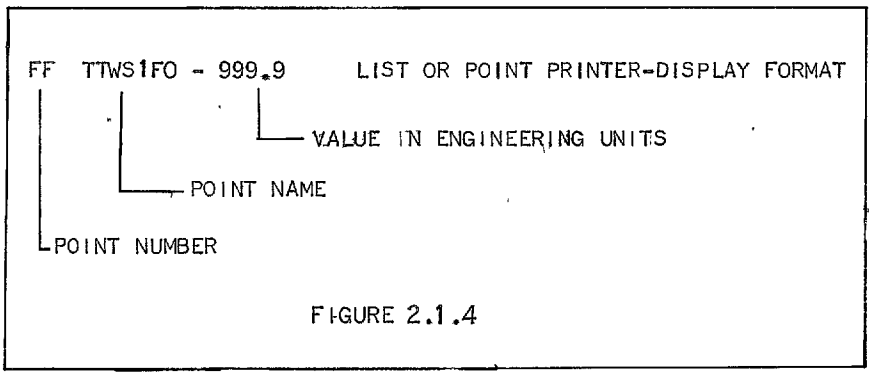


FIGURE 2.1.4

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS Azra Ehsan	GENERAL ELECTRIC	ELEMENTARY DIAGRAM	ITS APPLICATION NOTES
REV. 2	ISSUED Jan 25/79			FIRST MADE FOR		277A4936	
REV. 3	MADE BY AZRA EHSAN			I.C NO		CONT. ON SH. 25	SH. NO. 24

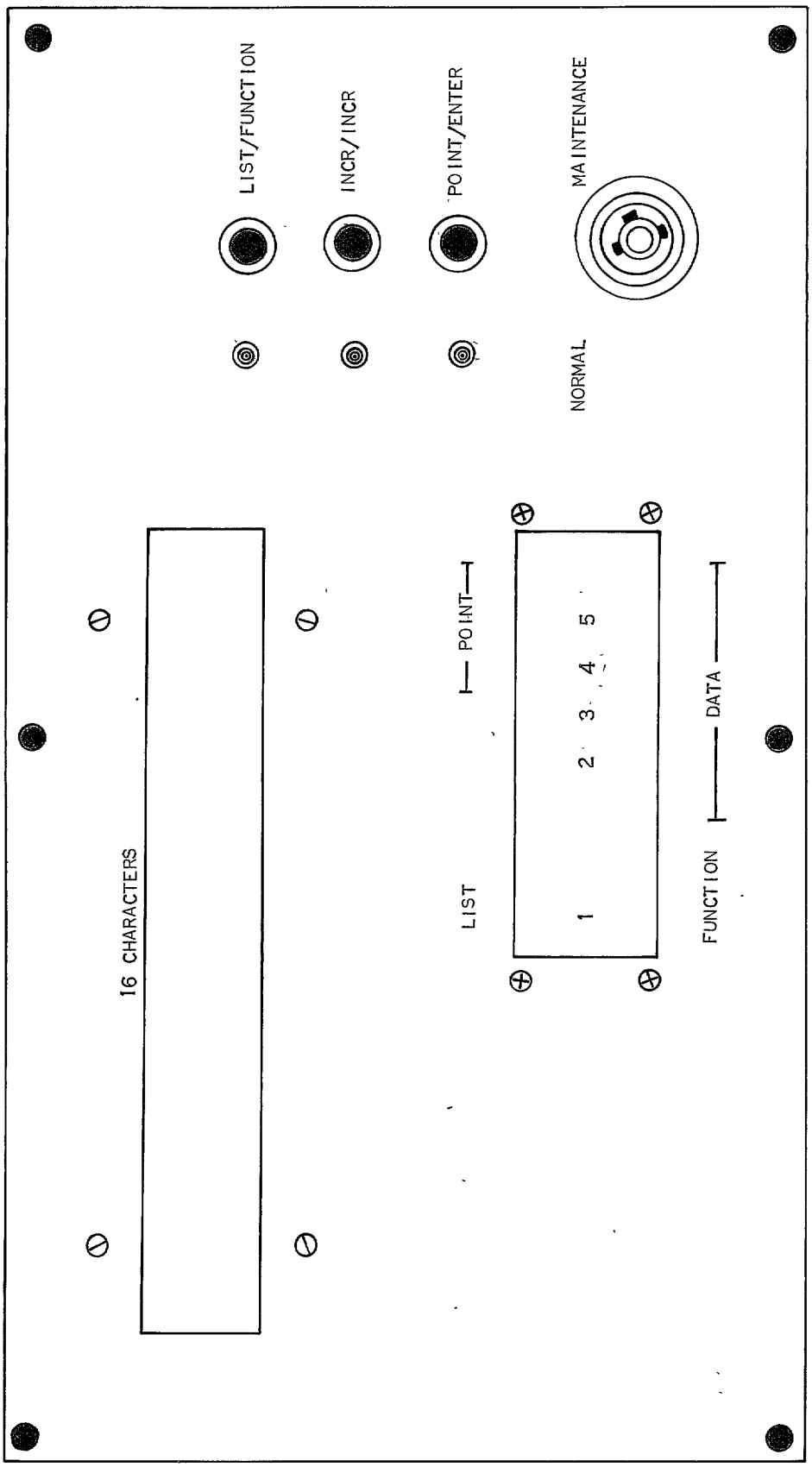


FIGURE 2.1.5

REV. 1	REV 4	REV. 5	PRINTS TO	APPROVALS Azra Ehsan	GENERAL ELECTRIC	ELEMENTARY DIAGRAM ITS APPLICATION NOTES
REV. 2	ISSUED Jan 25/79			FIRST MADE FOR		277A4936
REV. 3	MADE BY AZRA EHSAN			I.C. NO		CONT. ON SH. 26 SH. NO. 25

NORMAL MODE LIST THUMBWHEEL

LIST THUMBWHEEL	DESCRIPTION
0	NOT USED
1	BASIC OPERATING DATA
2	CONTROL EXHAUST TEMPERATURES
3	PROTECTIVE EXHAUST TEMPERATURES
4	WHEELSPACE TEMPERATURES
5	LUBE OIL TEMPERATURES (IF USED)
6	OPERATING DATA
7	REFERENCE DATA
8	(FUTURE)
9	(FUTURE)
A	PERIODIC LIST (OPTIONAL)
B	ALARM POINTS
C	CLOCK SET ROUTINE
D	(FUTURE)
E	HISTORICAL DATA
F	DISPLAY OF MEMORY CONTENTS

LIST 1: BASIC OPERATING DATA:

NHP	HIGH PRESSURE SET SPEED
TX	MEDIAN OF TX VECTOR
PCD	COMPRESSOR DISCHARGE PRESSURE
VCE	FUEL COMMAND

LIST 2: CONTROL EXHAUST TEMPERATURES:

TX	MEDIAN OF TX VECTOR
TTX-1	TURBINE EXHAUST DUCT THERMOCOUPLE #1
TTX-2	TURBINE EXHAUST DUCT THERMOCOUPLE #2
TTX-3	TURBINE EXHAUST DUCT THERMOCOUPLE #3
TTX-4	TURBINE EXHAUST DUCT THERMOCOUPLE #4
TTX-5	TURBINE EXHAUST DUCT THERMOCOUPLE #5
TTX-6	TURBINE EXHAUST DUCT THERMOCOUPLE #6
TTX-7	TURBINE EXHAUST DUCT THERMOCOUPLE #7
TTX-8	TURBINE EXHAUST DUCT THERMOCOUPLE #8
TTX-9	TURBINE EXHAUST DUCT THERMOCOUPLE #9
TTX-10	TURBINE EXHAUST DUCT THERMOCOUPLE #10
TTX-11	TURBINE EXHAUST DUCT THERMOCOUPLE #11
TTX-12	TURBINE EXHAUST DUCT THERMOCOUPLE #12
TTX-13	TURBINE EXHAUST DUCT THERMOCOUPLE #13

LIST 3: PROTECTIVE EXHAUST TEMPERATURES:

TX MEDIAN OF TX VECTOR
OTM MEDIAN OF OT VECTOR
OTA OVERTEMPERATURE CHANNEL A
OTB OVERTEMPERATURE CHANNEL B
OTC OVERTEMPERATURE CHANNEL C

LIST 4: WHEELSPACE TEMPERATURES:

WS-1F11	WHEELSPACE 1 FORWARD INNER #1
WS-1F12	WHEELSPACE 1 FORWARD INNER #2
WS-1FO1	WHEELSPACE 1 FORWARD OUTER #1
WS-1FO2	WHEELSPACE 1 FORWARD OUTER #2
WS-1AO1	WHEELSPACE 1 AFT OUTER #1
WS-1AO2	WHEELSPACE 1 AFT OUTER #2
WS-2FO1	WHEELSPACE 2 FORWARD OUTER #1
WS-2FO2	WHEELSPACE 2 FORWARD OUTER #2
WS-2AO1	WHEELSPACE 2 AFT OUTER #1
WS-2AO2	WHEELSPACE 2 AFT OUTER #2
WS-3FO1	WHEELSPACE 3 FORWARD OUTER #1
WS-3FO2	WHEELSPACE 3 FORWARD OUTER #2
WS-3AO1	WHEELSPACE 3 AFT OUTER #1
WS-3AO2	WHEELSPACE 3 AFT OUTER #2

LIST 5: LUBE OIL TEMPERATURES (IF USED):

POINTS MAY VARY PER JOB

LIST 6: OPERATING DATA:

VCE	-
FUEL FLOW	-
GVE	- GUIDE VANE POSITION REFERENCE SIGNAL
VIGV	- VARIABLE INLET GUIDE VANE POSITION
SPREAD 1	-
SPREAD 2	-

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV 4	REV. 5	PRINTS TO	APPROVALS Azra Ehsan	GENERAL ELECTRIC	ELEMENTARY DIAGRAM ITS APPLICATION NOTES
REV. 2	ISSUED Jan 25/79			FIRST MADE FOR		277A4936
REV. 3	MADE BY AZRA EHSAN			I.C NO		CONT. ON SH 28 SH. NO. 27

LIST 7: REFERENCE DATA

TXSET	- TEMPERATURE CONTROL REFERENCE
PCDBIAS	-
VCEBIAS	-
ALLOWABLE SPREAD	-
TCD-1	- COMPRESSOR DISCHARGE TEMPERATURE #1
TCD-2	- COMPRESSOR DISCHARGE TEMPERATURE #2
T-INLET-1	- COMPRESSOR INLET TEMPERATURES #1
T-INLET-2	- COMPRESSOR INLET TEMPERATURES #2
T-INLET-3	- COMPRESSOR INLET TEMPERATURES #3

LIST 8: (FUTURE)

LIST 9: (FUTURE)

LIST A: LIST A WILL DISPLAY PRESELECTED LIST POINTS ON THE ITS DISPLAY PANEL. 0-60 MINUTES SELECTABLE PERIOD MAY BE ENTERED FOR A PERIODIC PRINTOUT OF THE LIST A ON THE PRINTER. TIME BETWEEN 0 TO 60 MINUTES SET ON SWS(3) AND SWS(4) [DATA THUMBWHEELS] IS READ IN AND DISPLAYED WHEN THE POINT PUSH BUTTON IS ACTUATED. TIME SETTING OVER 60 MINUTES WILL BE IGNORED. ACTUATION OF POINT PUSH BUTTON PRIOR TO INCREMENT PUSH BUTTON WILL BE IGNORED. ZERO SETTING FOR RMIN WILL CAUSE CONTINOUS PRINTING.

1. DIAL "A" ON THW OR PRESS LIST PB WHEN THW = A
   (A) PRINT BLANK LINE.
   (B) DISPLAY AND PRINT DATE AND TIME.
   (C) DISPLAY AND PRINT ONE PASS THRU LIST.
   (D) CONTINUE TO DISPLAY ONLY UNTIL REPEAT TIME EXPIRES, THEN REPEAT FROM (A).

2. PRESS INC PB. DISPLAY PROMPTS:
   "SET REPEAT MINS"

3. DIAL INTERVAL IN MINUTES ON POINT THWS.
   PRESS POINT PB.
   DISPLAY SHOWS:
   "REPEAT MINS"=NN

4. PRESS LIST PB TO GO BACK TO LIST DISPLAY

5. PRESS INC PB TO GO BACK TO PROMPT MESSAGE

6. PRESS POINT PB TO ENTER DIFFERENT TIME

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS Azra Ehsan	GENERAL ELECTRIC	ELEMENTARY DIAGRAM ITS APPLICATION NOTES
REV. 2	ISSUED Jan 25/79			FIRST MADE FOR		2 7 7 A 4 9 3 6
REV. 3	MADE BY AZRA EHSAN			I C NO.		CONT. ON SH. 29 SH. NO. 28

LIST B: ALARM POINTS

LIST C: CLOCK AND CALENDAR

THIS IS CLOCK SETTING ROUTINE FOR THE TIME-OR-DAY CLOCK AND CALENDAR. LIST THUMBWHEEL IS SET AT 'C'. TIME AND DATE SETTINGS ARE ENTERED FROM SWS(1), SWS(2), SWS(3) AND SWS(4) [DATA THUMBWHEEL]. CLOCK ADJUSTMENT (INCREASE OR DECREASE) IN SECONDS PER DAY CAN BE MADE IN LIST "C". TO COMPENSATE FOR LONG TERM DRIFT IN ITS CLOCK ACTUATION OF INCREMENT PUSHBUTTON SELECTS ONE OF THE SIX CASES.

CASES:

CASE 0	DISPLAY CURRENT DATE AND TIME.
CASE 1	TIME
CASE 2	DATE
CASE 3	YEAR
CASE 4	CLK INCR
CASE 5	CLK DECR

ENTRY OF SETTING IS MADE UPON ACTUATION OF POINT PUSHBUTTON.

1. DIAL C ON LIST THUMBWHEEL TIME AND DATE ARE DISPLAYED.
2. PRESS INC PUSHBUTTON DISPLAY PROMPTS "SET HR: MIN"
3. SET HH MM ON THWS. PRESS POINT TO ENTER. REPEAT IF REQUIRED.
4. PRESS INC PB. DISPLAY PROMPTS "SET MO/DAY"
5. SET MM/DD ON THWS. PRESS POINT TO ENTER. REPEAT IF REQUIRED.
6. PRESS INC PB. DISPLAY PROMPTS "SET YEAR".
7. SET YEAR ON POINT THWS. PRESS POINT TO ENTER. REPEAT IF REQUIRED.
8. PRESS INC PB. DISPLAY PROMPTS "INCREASE SEC/DAY"
9. SET INCREASE IN SEC FOR CLOCK ON POINT THWS. PRESS POINT TB TO ENTER.
10. PRESS INC PB. DISPLAY PROMPTS "DECREASE SEC/DAY"
11. SET DECREASE IN SEC FOR CLOCK ON POINT THWS. PRESS POINT TB TO ENTER.
12. PRESS LIST PUSHBUTTON AT ANYTIME TO RETURN TO DATE AND TIME DISPLAY.
13. IF ANY THUMBWHEEL IS SET TO A THRU F OR BEYOND SPECIFIED LIMITS CHANGE IS IGNORED.

DATE AND TIME ARE DISPLAYED IN THE FOLLOWING FORMAT:

27NOV78 12:45:30

FIGURE 2.1.6

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS Azra Ehsan	GENERAL ELECTRIC	ELEMENTARY DIAGRAM ITS APPLICATIONS NOTES
REV 2	ISSUED Jan 25/79			FIRST MADE FOR		277A4936
REV 3	MADE BY AZRA EHSAN			I.C. NO.		CONT. ON SH. 30 SH. NO. 29

LIST E: HISTORICAL RECORD

IF "E" IS DIALED ON THE LIST THUMBWHEEL OF THE DISPLAY PANEL, DATA WILL BE CAPTURED WITH INTERRUPTS DISABLED. 5 BLOCKS OF DATA OF VARYING AGE WILL BE PRINTED WITH SINGLE BLANK LINE BETWEEN EACH BLOCK. THE VALUES OF THE FOLLOWING WILL BE DISPLAYED IN EACH BLOCK

EPOCH (TIME)
AGE
ALARM1
ALARM2
NHP
VCE
VIGV
PCD
FFN/D
P5
ACOM
3 OTS
13 TXS

LIST F: MEMORY DISPLAY

IF LIST PB IS ACTUATED NEW DATA WILL BE CAPTURED AND NEW LIST WILL BE PRINTED. OTHERWISE LIST WILL BE REPEATED WITH OLD DATA. ACTUATION OF INCR OR POINT PB WILL BE IGNORED.

LIST "F" IS USED FOR HEXADECIMAL DISPLAY OF MEMORY CONTENTS IN NORMAL MODE. MEMORY ADDRESS IS READ FROM DATA THUMBWHEELS EACH TIME LIST BUTTON IS PRESSED. ADDRESS MAY BE INCREMENTED USING INC BUTTON. DISPLAY FORMAT IS HEX-BINARY. MEMORY-MAPPED I/O. ADDRESSES ARE NOT DISPLAYED. IT IS NOT POSSIBLE TO ENTER DATA INTO MEMORY IN NORMAL MODE.

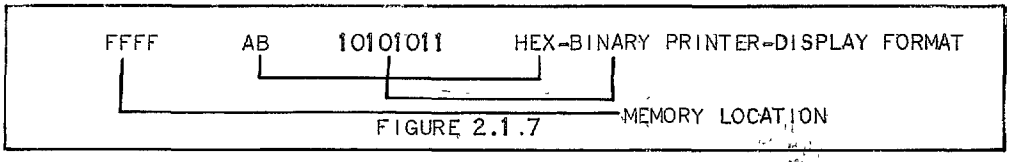


FIGURE 2.1.7

B. POINT DISPLAY - THIS MODE IS ENTERED FROM LIST DISPLAY MODE BY PRESSING POINT PUSHBUTTON. PANEL POWERS UP IN THIS MODE IF KEY IS IN NORMAL POSITION.

1. ENTER DESIRED POINT NUMBER ON POINT THUMBWHEELS. POINT NUMBER IS HEXADECIMAL.

2. PRESS POINT PUSHBUTTON. POINT LIGHT COMES ON. LIST LIGHT GOES OFF. DESIRED POINT IS DISPLAYED AND UPDATED EVERY 2 SECONDS.

3. POINT THUMBWHEELS CAN BE CHANGED AT ANY TIME. NEW POINT WILL BE DISPLAYED WITHIN ONE-HALF SECOND.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV 1	REV. 4	REV 5	PRINTS TO	APPROVALS Azra Ehsan	GENERAL ELECTRIC	ELEMENTARY DIAGRAM ITS APPLICATION NOTES
REV. 2	ISSUED Jan 25/79			FIRST MADE FOR		2 7 7 A 4 9 3 6
REV. 3	MADE BY AZRA EHSAN			I.C NO		CONT. ON SH. 31 SH. NO. 30

4. PRESSING THE INC BUTTON CAUSES THE DISPLAYED POINT NUMBER TO BE INCREMENTED, AND THE VALUE OF THE NEXT POINT TO BE DISPLAYED. INC LIGHTS COMES ON, AND REMAINS ON. POINT LIGHT GOES OFF. PUSHING INC AGAIN CAUSES THE POINT NUMBER TO BE INCREMENTED BY ONE EACH TIME. POINT THUMBWHEELS ARE IGNORED UNTIL POINT BUTTON IS PUSHED AGAIN. INC BACKLIGHT GOES OFF WHEN POINT BUTTON IS PRESSED (OR WHEN LIST BUTTON OR KEYSWITCH ARE USED)..

5. LIST THUMBWHEEL IS IGNORED IN POINT DISPLAY MODE.

C. PRINTER ACTION - PRINTER POWER IS TURNED ON BY A TOGGLE SWITCH MOUNTED ON THE PRINTER. A PILOT LIGHT INDICATES POWER ON.

1. RAISE PRINT TOGGLE SWITCH. PILOT LIGHT COMES ON. THIS CAN BE DONE AT ANY TIME.

2. EACH TIME POINT OR INC BUTTONS ARE PRESSED, ONE LINE IS PRINTED IDENTICAL TO DISPLAY CONTENTS. OTHERWISE, PRINT-OUT OCCURS EVERY 2 SECONDS WHEN DISPLAY IS UPDATED. POINT BUTTON IS IGNORED WHEN DISPLAYING LIST F (MEMORY CONTENTS).

3. EACH TIME LIST BUTTON IS PRESSED A BLANK LINE IS PRINTED, FOLLOWED BY ONE LINE FOR EACH POINT IN THE LIST.
WHEN DISPLAYING LIST F (MEMORY CONTENTS) PRESSING LIST BUTTON CAUSES ONLY THE CURRENT LINE TO BE PRINTED.

4. WHEN IN LIST C, PRINTER IS DISABLED.
WHEN IN LIST E, LIST IS PRINTED EVERY 1/2 SECOND.
WHEN IN LIST "A", PRINT INTERVAL IS DETERMINED BY STORED "REPEAT MINUTES" (SEE PAGE 28).

2. MAINTENANCE OPERATION - THIS MODE IS SELECTED BY PUTTING THE KEY SWITCH IN MAINTENANCE POSITION. THE KEY SWITCH IS IGNORED (IE. PANEL REMAINS IN NORMAL MODE) IF THE TURBINE IS RUNNING OR IN A START-UP SEQUENCE. THE TURBINE WILL NOT START IF THE PANEL IS LEFT IN MAINTENANCE MODE. SWITCHING FROM MAINTENANCE TO NORMAL MODE CAUSES RE-INITIALIZATION OF ITS.

IN MAINTENANCE MODE, THE FUNCTION AND DATA THUMBWHEELS ARE READ ONLY WHEN A FUNCTION BUTTON IS PRESSED. CHANGING THUMBWHEEL SETTINGS WITHOUT PRESSING A FUNCTION BUTTON HAS NO EFFECT.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS Azra Ehsan	GENERAL ELECTRIC	ELEMENTARY DIAGRAM ITS APPLICATION NOTES
REV. 2	ISSUED Jan 25/79			FIRST MADE FOR		277 A4936.
REV. 3	MADE BY AZRA EHSAN			I.C. NO		CONT. ON SH. 32 SH. NO. 31

MAINTENANCE MODE FUNCTION THUMBWHEEL

FUNCTION CODE	FUNCTION
0	MEMORY WRITE (HEX-BINARY)
1	MEMORY READ (HEX-BINARY)
2	MEMORY PRINT (4-BYTE HEX)
3	I/O READ (DAS ONLY)
4	I/O WRITE (DAS ONLY)
5	
6	
7	
8	
9	OTHER MAINTENANCE FUNCTIONS (FUTURE)
A	
B	
C	
D	
E	
F	

A. MEMORY WRITE - CODE 0. THIS MODE IS IDENTICAL TO MEMORY READ, EXCEPT THAT DATA MAY BE ENTERED INTO ANY WRITABLE MEMORY LOCATION OR MEMORY-MAPPED I/O PORT.

1. SET FUNCTION CODE TO 0.

2. SET DESIRED ADDRESS ON DATA THUMBWHEELS.

3. PRESS FUNCTION BUTTON. DESIRED ADDRESS AND CONTENTS APPEAR ON THE DISPLAY IN HEX-BINARY FORMAT.

4. ADDRESS MAY BE INCREMENTED BY PRESSING INC BUTTON.

5. ADDRESS MAY BE CHANGED AT ANY TIME BY SETTING A NEW ADDRESS ON DATA THUMBWHEELS AND PRESSING FUNCTION BUTTON.

6. DATA MAY BE ENTERED, ONE BYTE AT A TIME, INTO ANY WRITABLE LOCATION BY SETTING THE DESIRED DATA ON THE TWO LEAST SIGNIFICANT DATA THUMBWHEELS, AND PRESSING THE ENTER BUTTON.

7. DISPLAYING AN INPUT DEVICE REGISTER MAY CAUSE STATUS FLAGS TO BE CHANGED. THIS MAY RESULT IN UNEXPECTED DEVICE ACTION.

8. DATA MAY BE WRITTEN DIRECTLY TO MEMORY-MAPPED I/O DEVICE REGISTERS USING THIS MODE. DEVICE ACTION MAY RESULT.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS Azra Ehsan	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED Jan 25/79			FIRST MADE FOR		ITS APPLICATION NOTES
REV. 3	MADE BY AZRA EHSAN			I.C. NO.		277A4936
						CONT. ON SH. 33 SH. NO. 32

9. ATTEMPTS TO WRITE TO ROM LOCATIONS WILL BE EXECUTED, BUT WILL NOT CHANGE ROM CONTENTS. ATTEMPTS TO READ OR WRITE TO NON-EXISTENT MEMORY LOCATIONS WILL BE EXECUTED. THE DISPLAY WILL SHOW THE SAME DATA AS SEEN BY THE PROCESSOR. THIS APPLIES ALSO TO I/O PORTS, WHICH MAY BE UNI-DIRECTIONAL (IE. NO INPUT PORT ON AN OUTPUT PORT ADDRESS OR VICE-VERSA).

B. MEMORY READ - CODE 1 THIS MODE ALLOWS THE OPERATOR TO DISPLAY THE CONTENTS OF ANY MEMORY LOCATION OR MEMORY-MAPPED I/O DEVICE REGISTER. DATA MAY NOT BE WRITTEN TO MEMORY OR I/O PORTS IN THIS MODE.

1. SET FUNCTION CODE TO 1.

2. SET DESIRED ADDRESS ON DATA THUMBWHEELS.

3. PRESS FUNCTION BUTTON. DESIRED ADDRESS AND CONTENTS APPEAR ON THE DISPLAY IN HEX-BINARY FORMAT.

4. ADDRESS MAY BE INCREMENTED BY PRESSING INC BUTTON.

5. ADDRESS MAY BE CHANGED AT ANY TIME BY SETTING A NEW ADDRESS ON DATA THUMBWHEELS AND PRESSING FUNCTION BUTTON.

6. DISPLAYING AN INPUT DEVICE REGISTER MAY CAUSE STATUS FLAGS TO BE CHANGED. THIS MAY RESULT IN UNEXPECTED DEVICE ACTION.

C. MEMORY PRINT - CODE 2 THIS MODE ALLOWS CONTINUOUS PRINT-OUT OF MEMORY IN HEXADECIMAL FORMAT, FOUR BYTES PER LINE.

1. SET FUNCTION CODE TO 2.

2. SET DESIRED ADDRESS ON DATA SWITCHES.

3. PRESS FUNCTION BUTTON. DESIRED ADDRESS AND CONTENTS OF FOUR CONSECUTIVE BYTES, BEGINNING WITH ADDRESSED BYTE, ARE DISPLAYED, EVERY HALF SECOND.

4. PRESSING INC BUTTON WILL CAUSE ADDRESS TO BE INCREMENTED BY FOUR AND FOUR NEW BYTES OF DATA WILL BE DISPLAYED.

5. PRESSING ENTER WILL CAUSE AUTOMATIC INCREMENTING AND PRINTING OF ONE LINE EVERY HALF-SECOND. PRINT BUTTON MUST HAVE BEEN PRESSED TO ENABLE PRINTING. ALTERNATE PRESSING OF ENTER BUTTON STOPS AND STARTS AUTO INCREMENT.

FFFF AB CD EF 01 4-BYTE HEX

FIGURE 2.1.8

D. I/O READ - CODE 4

IDENTICAL TO CODE 1 EXCEPT THAT I/O ADDRESS SPACE IS ACCESSED. TWO MS DIGITS OF ADDRESS ARE BLANKED ON DISPLAY, IGNORED ON THUMB-WHEELS.

E. I/O WRITE - CODE 3

IDENTICAL TO CODE 0, EXCEPT THAT I/O SPACE IS ACCESSED. TWO MS DIGITS ARE BLANKED ON DISPLAY AND PRINTER, IGNORED ON THUMBWHEELS.

2.2.0 ANALOG INPUT RACK/STICK

THE ANALOG INPUT RACK IS USED IN THE ITS TO CONNECT THERMOCOUPLES FROM VARIOUS POINTS IN THE TURBINE (EXCEPT FOR OVERTEMPERATURE THERMOCOUPLES WHICH ARE CONNECTED TO THE NOTA MODULE).

ANY TYPE OF THERMOCOUPLE MAY BE USED PROVIDED THE CORRESPONDING COLD JUNCTION COMPENSATION SOFTWARE IS AVAILABLE.

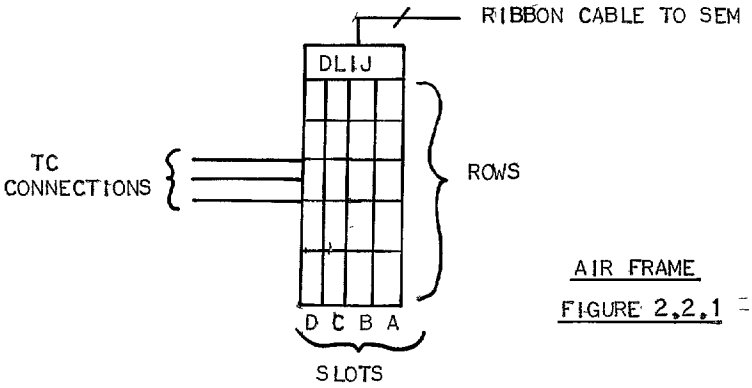
AIR BASICALLY CONSISTS OF A FRAME WITH CARD RECEPTACLES AND BACK PLANE WIRING. THE FRAME IS DESIGNED WITH GUIDE RAILS TO ACCOMODATE FUNCTIONAL CARDS.

THE NUMBER OF THERMOCOUPLES THAT CAN BE CONNECTED DEPENDS ON THE SIZE OF AIR AND THE FUNCTIONAL CARDS IN THE AIR. AIR SIZES AVAILABLE ARE AS FOLLOWS

1C4516CA104 - 2 CHANNELS	32 POINTS	27 TC TERMINATIONS AVAILABLE
1C4516CA108 - 4 CHANNELS	64 POINTS	54 TC TERMINATIONS AVAILABLE
1C4516CA116 - 8 CHANNELS	128 POINTS	82 TC TERMINATIONS AVAILABLE

NOTE: GENERALLY THERE ARE 16 POINTS PER CHANNEL.
2 POINTS PER CHANNEL ARE PRE-ASSIGNED FOR COLD JUNCTION RTD
1 POINT PER AMPLIFIER CARD IS RESERVED FOR SHORTED INPUT USED FOR OFFSET COMPENSATION.

THE PHYSICAL LOCATION OF THE AIR IN THE CONTROL CABINET MAY VARY. TC CONNECTIONS ARE MADE AT THE TERMINALS ON THE FRONT OF THE ATCA CARD ASSEMBLY LOCATED IN THE AIR.

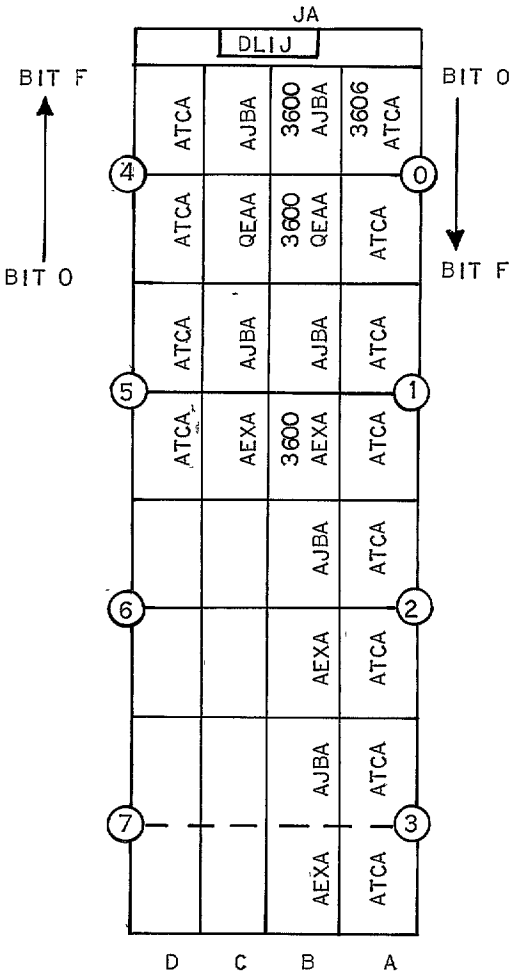


AIR FRAME

FIGURE 2.2.1

CONNECTIONS TO THE SEM ARE MADE WITH A 34 PIN RIBBON CABLE CONNECTOR LOCATED ON THE 1C3600 DLIJ CARD LOCATED AT THE UPPER END OF AIR.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS Azra Ehsan	GENERAL ELECTRIC	ELEMENTARY DIAGRAM ITS APPLICATION NOTES
REV. 2	ISSUED Jan 25/79			FIRST MADE FOR		277A4936
REV. 3	MADE BY AZRA EHSAN			I.C. NO.		CONT. ON SH. 36 SH. NO. 35



AIR CARD LAYOUT

FIGURE 2.2.2

REV 1	REV. 4	REV 5	PRINTS TO	APPROVALS Azra Ehsan	GENERAL ELECTRIC	ELEMENTARY DIAGRAM ITS APPLICATION NOTES
REV. 2	ISSUED Jan 25/79			FIRST MADE FOR		277A4936
REV 3	MADE BY AZRA EHSAN			I.C. NO		CONT. ON SH. 37 SH. NO 36

THE FOLLOWING CONNECTIONS ARE MADE TO SEM;

FROM SEM	P15	PS3 POWER SUPPLY	
	ACOM		
	N15		
	ENSW	NADC CARD	
	AO TO A6		
TO SEM	CHOP TO CH5P	NADC CARD (A/D INPUTS)	
	CHON TO CH5N		

POWER IN THE AIR IS FED FROM THE PS3 POWER SUPPLY.

FUNCTIONS OF THE ANALOG INPUT RACK IN ITS

1. TERMINATE THE INPUT THERMOCOUPLE SIGNALS
2. SIGNAL CONDITION AND FILTER THESE SIGNALS
3. SELECT ON A TIME SHARED BASIS, ONE POINT OUT OF 16 POINTS ON EACH CHANNEL, WITH THE EXPAND CAPABILITY, MORE THAN 16 POINTS CAN BE HANDLED BY A SINGLE AMPLIFIER (TO REDUCE GAIN AND OFFSET ERRORS)
4. AMPLIFY THE INPUT SIGNAL SO THAT IT APPEARS AS A NORMALIZED VOLTAGE FOR THE A/D CONVERTER. NORMALLY ±10V WILL BE MAXIMUM LEVEL.

THE AIR IS VERTICALLY DIVIDED BETWEEN SLOTS A AND B AS ONE SIDE AND SLOTS C AND D AS THE OTHER SIDE. ONLY ONE AMPLIFIER CARD (QEAA) IS USED PER EACH SIDE OF THE AIR. EXPANDER CARDS (AEXA) ARE USED FOR ADDITIONAL SIGNALS FOR EACH SIDE OF AIR. THE USE OF SINGLE AMPLIFIER FOR EACH SIDE OF AIR AIDS IN MINIMIZING ERRORS DUE TO DIFFERENT GAINS AND OFFSETS.

TWO ATCA CARDS (THERMOCOUPLE CONNECTIONS) ARE ASSIGNED TO EACH OF THE AMPLIFIER OR EXPANDER CARD IN THE AIR. THIS IS CALLED A "CHANNEL". EACH CHANNEL CONSISTS OF TWO ATCA CARDS (16 POINTS) AND EITHER AN AMPLIFIER (QEAA) OR EXPANDER (AEXA) CARD. EACH ATCA CARD CONSISTS OF 8 POINTS. FIRST POINT ON ALL ATCA CARDS IS RESERVED FOR COLD JUNCTION COMPENSATION INPUTS. ALSO FOR EACH SIDE OF AIR, ONE POINT ON ATCA IS RESERVED FOR A SHORTED INPUT FOR OFFSET COMPENSATION CALCULATIONS DONE BY THE SOFTWARE.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS Azra Ehsan	GENERAL ELECTRIC	ELEMENTARY DIAGRAM ITS APPLICATION NOTES
REV. 2	ISSUED Jan 25/79			FIRST MADE FOR		277A4936
REV. 3	MADE BY AZRA EHSAN			I.C. NO.		CONT. ON SH 38 SH. NO. 37

2.2.0 AIR SCANNING SYSTEM

THERMOCOUPLE INPUTS ON THE AIR ARE SCANNED PERIODICALLY BY THE CPU. THE CPU COMMUNICATES DIRECTLY WITH A MEMORY-MAPPED I/O PORT ON THE NADC CARD (SEM) TO SELECT A SINGLE POINT IN THE AIR. THE LEAST SIGNIFICANT SEVEN BITS OF PORT C ON THE NADC ARE USED AS ADDRESS LINES TO AIR.

THE PROCESS OF SELECTING A SINGLE POINT MAY BE SIMPLY EXPLAINED AS FOLLOWS. REFER TO FIGURE 2.2.3. THE UPPER THREE BITS OF THE ADDRESS LINES ($A_4A_5A_6$) ARE COMPARED WITH THE THREE 'ADDR' STRAPPINGS OF THE QEAA AND AEXA CARDS IN AIR. THUS A SINGLE CARD (EITHER QEAA OR AEXA) IS SELECTED. THE SELECTED CARD WILL GENERATE A CHANNEL ENABLE SIGNAL "OPERM" WHICH WILL ENABLE THE TWO ATCA CARDS ASSIGNED TO THE SELECTED QEAA OR AEXA CARD. NOW OF THE TWO ATCA CARDS ONE IS TO BE SELECTED. THIS IS DONE BY COMPARING A3 OF THE ADDRESS LINES WITH THE STRAPPING 'ADDR' OF THE ATCA CARDS, (SINCE PER CHANNEL ONE ATCA 'ADDR' IS TIED TO GROUND AND THE OTHER ATCA 'ADDR' IS LEFT OPEN). WITH ONE OF THE TWO ATCA CARDS SELECTED, THE TASK IS OF SELECTING ONE OF THE EIGHT POINTS ON ATCA. THIS IS DONE BY A THREE TO EIGHT DECODER ON THE ATCA CARD. THE LOWEST THREE ADDRESS LINES A_0, A_1, A_2 ARE DECODED TO SELECT ONE OF THE EIGHT POINTS.

THE CONDITIONED SIGNAL OF THE POINT SELECTED BY THE ABOVE TECHNIQUE IS PASSED THRU THE AUXILLARY SELECT SWITCH LOCATED ON THE SELECTED QEAA OR AEXA CARDS (WHICH WOULD BE CLOSED BECAUSE A_3-A_6 WOULD HAVE SELECTED THIS CARD). THROUGH THE AUXILLARY SELECT SWITCH THIS SIGNAL IS FED TO THE INSTRUMENTATION AMPLIFIER ON THE QEAA CARD.

THE OUTPUT SIGNAL OF THE AMPLIFIER IS DEMULTIPLEXED INTO THE PROPER CHANNEL (CHANNELS 0 TO CHANNEL 5) ON THE NADC CARD ($A_3A_4A_5$ ARE ALSO USED ON NADC FOR DEMULTIPLEXING).

ENSW GOES TO A "ZERO" FOR 9.1 MICROSECONDS AT THE START OF EACH ADDRESSING CYCLE TO THE NADC CARD. THE FUNCTION IS TO OPEN ALL ANALOG INPUT SCANNER MULTIPLEX SWITCHES PRIOR TO CHANGING ADDRESS TO PREVENT SWITCH OVERLAP.

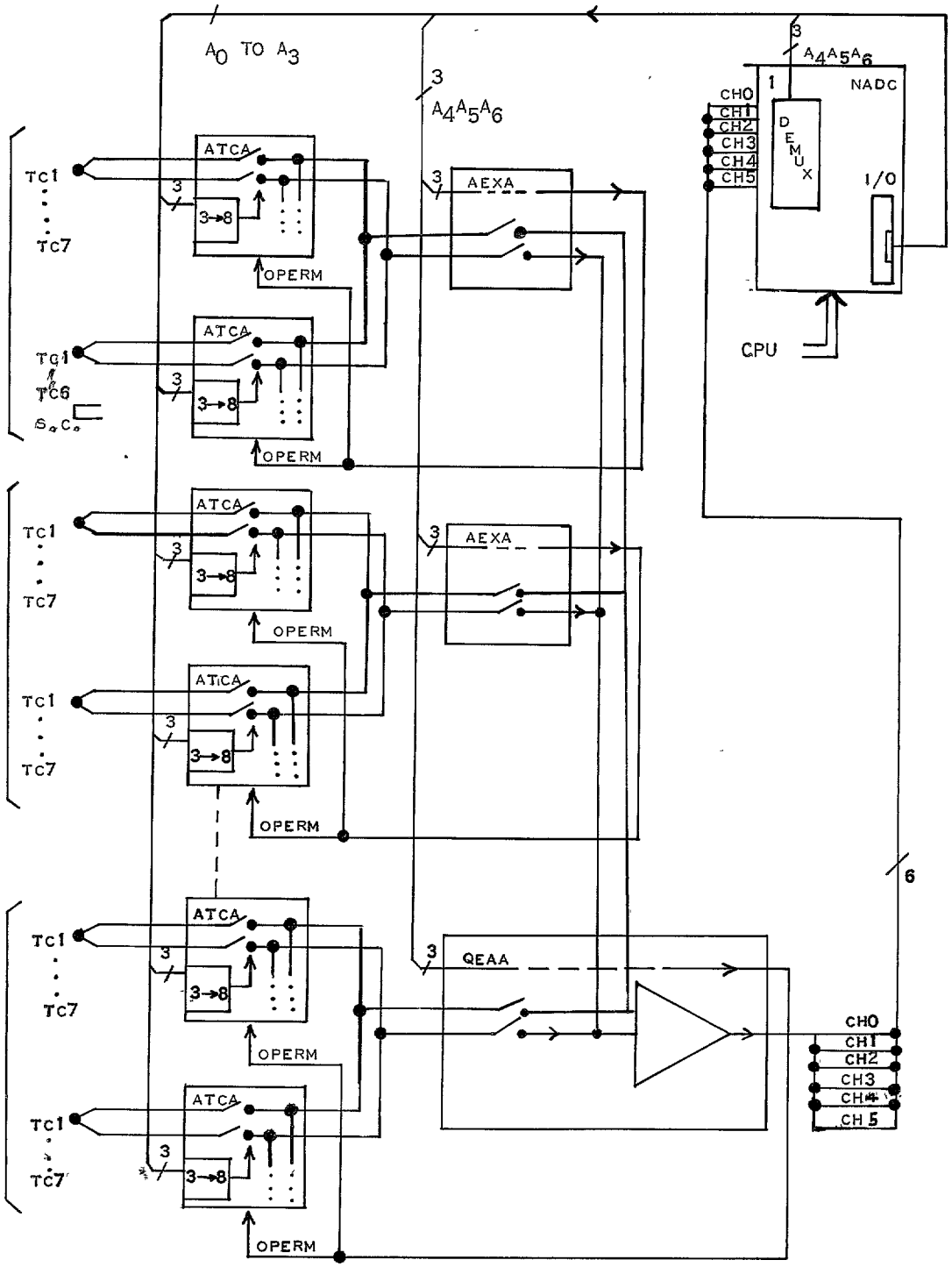
THE CARDS ASSOCIATED WITH THE PRECEEDING FUNCTIONS IN THE AIR ARE THE FOLLOWING:

IC3606 ATCA	THERMOCOUPLE CONDITIONING
IC3600 AJBA	JUMPER BOARD
IC3600 QEAA	EXPANDABLE AMPLIFIER
IC3600 AEXA	EXPANDER CARD

REFER TO SYSTEM ELEMENTARY DIAGRAMS FOR INPUT/OUTPUT PIN CONNECTIONS AND SYMBOLIC DIAGRAM OF THESE CARDS.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS Azra Ehsan	GENERAL ELECTRIC	ELEMENTARY DIAGRAM ITS APPLICATION NOTES
REV. 2	ISSUED Jan 25/79			FIRST MADE FOR		277A4936
REV. 3	MADE BY AZRA EHSAN			I.C. NO		CONT. ON SH. 39 SH. NO. 38

FORM 1C 2390 12-74



AIR SCANNING SYSTEM

FIGURE 2.2.3

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS Azra Ehsan	GENERAL ELECTRIC	ELEMENTARY DIAGRAM ITS APPLICATION NOTES
REV. 2	ISSUED Jan 25/79			FIRST MADE FOR		277A4936
REV. 3	MADE BY AZRA EHSAN			I.C. NO.		CONT. ON SH 40 SH NO. 39

2.2.1-1 IC3606 ATCA THERMOCOUPLE CONDITIONING CARD

THIS CARD IS USED WITH THE EXPANDABLE DIFFERENTIAL AMPLIFIER IC3600QEAA OR SIMILAR TYPE CARD. IN AN ANALOG I/O RACK.

IT PROVIDES A COLD JUNCTION TERMINATION FOR AS MANY AS SEVEN THERMOCOUPLE INPUTS. IT ALSO CONTAINS A RTD TEMPERATURE SENSOR EMBEDED IN THE COLD JUNCTION TO MEASURE THE COLD JUNCTION TEMPERATURE.

THE SEVEN THERMOCOUPLE INPUTS AND THE COLD JUNCTION TEMPERATURE SENSOR ARE MULTIPLEXED TO A COMMON DIFFERENTIAL OUTPUT BUSS.
INPUT ADDRESS INFORMATION IS RECEIVED FROM THE SEM (NADC CARD). TWO OF THESE CARDS MAY BE USED PER CHANNEL OF AIR TO PROVIDE 14 ADDRESSABLE THERMOCOUPLE INPUT POINTS.

EACH THERMOCOUPLE INPUT HAS A TWO POLE PASSIVE FILTER WHICH PROVIDES 40 DB OF ATTENUATION AT 60 HZ. TO NORMAL MODE SIGNALS.

THE INPUTS ARE SCANNED BY THE CPU IN THE SEM.

CARD FUNCTIONS

A. COLD JUNCTION TEMPERATURE SENSOR- THIS CIRCUIT CONSISTS OF A RESISTOR BRIDGE NETWORK OF WHICH THE RTD IS ONE LEG OF THE BRIDGE. THE BRIDGE OUTPUT GRADIENT IS -22.481 $\mu V/^{o}F$. REFERENCED TO $77^{o}F$ (I.E. OUTPUT IS ZERO UV AT $77^{o}F$). THE BRIDGE CIRCUIT IS THE FIRST ADDRESSABLE POINT ON THE CARD (POINT 0). IF TWO CARDS USED IN A ROW POINT 0 OF THE SECOND CARD WOULD BE THE 8TH ADDRESSABLE POINT ON THE ROW.

B. THERMOCOUPLE CONDITIONING - EACH THERMOCOUPLE INPUT HAS ASSOCIATED WITH IT A LOW PASS FILTER WHICH PROVIDES 40 DB OF ATTENTUATION TO 60 HZ NORMAL MODE SIGNALS. THE THERMOCOUPLE IS RAISED TO A COMMON MODE VOLTAGE OF ABOUT 0.5 VOLT. THIS IS USEFUL AS AN OPEN CIRCUIT DETECTOR SINCE AN OPEN THERMOCOUPLE WOULD DRIVE THE EXPANDABLE DIFFERENTIAL AMPLIFIER INTO SATURATION. THEREFORE ANY UNUSED THERMOCOUPLE INPUTS SHOULD BE SHORTED WITH A JUMPER WIRE, AT THE ATCA TERMINAL BOARD.

C. ADDRESS DECODING - THE FOUR LEAST SIGNIFICANT BITS OF THE ADDRESS ARE USED TO DECODE POINTS "0" TO "7" ON THIS CARD AS WELL AS POINTS "8" TO "15" ON A SECOND ATCA CARD OF THE SAME CHANNEL IN AIR. SELECTING OF THE FIRST EIGHT OR SECOND EIGHT POINTS IS DETERMINED BY FIXED STRAPPING OF THE ADDRESS (PIN 40) ON THE AIR. A STRAP TO ACOM ON PIN 40 SELECTS POINT "0" THROUGH POINT "7". AN OPEN SELECTS POINTS "8" THROUGH "15". A "OPERM" SIGNAL WHEN LOGIC 1 WILL INHIBIT ADDRESSING OF ANY POINT OF THIS CARD.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS Azra Ehsan	GENERAL ELECTRIC	ELEMENTARY DIAGRAM IITS APPLICATION NOTES
REV. 2	ISSUED Jan 25/79			FIRST MADE FOR		277A4936
REV. 3	MADE BY AZRA EHSAN			I.C NO.		CONT. ON SH. 41 SH. NO. 40

ATCA INPUT TERMINAL BLOCK LABELING

P1,N1	POINT #0	THERMOCOUPLE INPUT
P2,N2	POINT #1	THERMOCOUPLE INPUT
P3,N3	POINT #2	THERMOCOUPLE INPUT
P4,N4	POINT #3	THERMOCOUPLE INPUT
P5,N5	POINT #4	THERMOCOUPLE INPUT
P6,N6	POINT #5	THERMOCOUPLE INPUT
P7,N7	POINT #6	THERMOCOUPLE INPUT
P8,N8	POINT #7	THERMOCOUPLE INPUT

COM CONNECTED TO ACOM (19)

IMPORTANT - JUMPER UNUSED INPUTS P TO N.

INPUTS

A0I	(45)	LEAST SIGNIFICANT BIT OF ADDRESS - OV "0" = 15V = "1"
A1I	(44)	LEAST SIGNIFICANT BIT OF ADDRESS - OV "0" = 15V = "1"
A2I	(43)	LEAST SIGNIFICANT BIT OF ADDRESS - OV "0" = 15V = "1"
A3I	(42)	MOST SIGNIFICANT BIT OF ADDRESS - OV "0" = 15V = "1"

ADDR (40) ADDRESS STRAPPING - STRAPPED TO ACOMI FOR ADDRESS 0-7. OPEN FOR ADDRESS 8-15.

ENSWI ENABLE SWITCH - WHEN A ZERO MAKES ALL DECODER OUTPUTS FALSE. USED TO INSURE NO SWITCH OVERLAPPING AT THE MULTIPLEXER.

OPERM (25) ADDRESSING PERMISSIVE - ENABLES ADDRESSING POINTS ON THIS CARD WHEN LOGIC 0. A LOGIC 1 INHIBITS ALL POINTS IN THIS CHANNEL.

OUTPUTS

P+ (3,49)
P- (4,48)

MULTIPLEXED DIFFERENTIAL OUTPUT BUSS TO DIFFERENTIAL AMPLIFIER CARD. ABOUT 30MV FULL SCALE SIGNAL P+ TO P-.

BUSSES

P15AI - 15V. 45MA MAX.
N15AI - -15V. 25MA MAX.

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REV. 2	ISSUED Jan 25/79			FIRST MADE FOR		277A4936
REV. 3	MADE BY AZRA EHSAN			I.C. NO.		CONT. ON SH. 42 SH. NO. 41

2.2.1-2 IC3600 AJBA JUMPER CARD

THIS CARD IS INSERTED IN SLOTS B OR C OF THE AIR TO CONNECT THE POWER (±15 VOLT, ACOM), ADDRESS (A_0 TO A_6) AND CONTROL ENSW) SIGNALS. IT ALSO PROVIDES LOCALIZED FILTERING OF POWER BUSSES AND RF BY-PASSING TO EARTH GROUND.

2.2.1-3 IC3600 QEAA EXPANDABLE AMPLIFIER CARD

THIS CARD CONSISTS OF AN AUXILLARY CHANNEL DECODER AND HIGH GAIN INSTRUMENTATION TYPE AMPLIFIER. THIS CARD IS FILTERED TO MINIMIZE THE EFFECTS OF RADIO FREQUENCY INTERFERENCE.(RF1).

INPUTS TO THIS CARD ARE NORMALLY RECEIVED FROM AN IC3606ATCA OR SIMILAR TYPE CARD.

THE AUXILLARY CHANNEL DECODER IS IMPLEMENTED SUCH THAT WHEN THE "I" AND "ADDR" INPUTS ARE MATCHED ON A BIT FOR BIT BASIS THE "AUXILLARY CHANNEL SWITCH" WILL CLOSE. THE AUXILLARY CHANNEL SWITCH WILL ALSO CLOSE WHEN ALL ZEROS APPEAR AT THE "I" INPUTS REGARDLESS OF THE ADDR STRAPPING. THE ADDRESS STRAPPING IS FIXED ON THE AIR, AND THE AUXILLARY SWITCH CLOSURE IS DETERMINED BY THE AMPLIFIER CARD "ROW" LOCATION IN THE RACK, OR BY AN ASSOCIATED EXPANDER CARD LOCATION IN THE RACK.

HIGH GAIN INSTRUMENTATION AMPLIFIER

THE INPUT STAGE OF THE INSTRUMENTATION AMPLIFIER IS MADE UP OF TWO LOW DRIFT AMPLIFIERS. THE AMPLIFIERS ARE IN A BALANCED CONFIGURATION. DIODES PROVIDE OVER VOLTAGE PROTECTION BY CLAMPING THE SIGNAL INPUTS TO THE POWER SUPPLY BUSSES. CAPACITORS ARE USED TO "HOLD" THE AMPLIFIER WHILE THE AUXILLARY CHANNEL SWITCH IS OPEN.

THE OUTPUTS OF THE INPUT STAGE DRIVE A STANDARD DIFFERENTIAL INPUT AMPLIFIER WHICH HAS A SINGLE ENDED OUTPUT CH+(14) REFERRED TO ACOM CH-(15).

ADJUSTMENTS

NO FIELD ADJUSTMENTS SHOULD BE MADE ON THIS CARD. ALL POTS ARE SEALED AT THE FACTORY.

PERFORMANCE CHARACTERISTICS

LONG TERM DRIFT -0.6MV PER MONTH. (TA CONST).

NORMAL MODE 60HZ ATTENUATION - 40DB MIN

COMMON MODE REJECTION RATIO. 80DB MIN

MAXIMUM COMMON MODE VOLTAGE - OPERATION ± 13 VOLTS ON ANY ONE INPUT - ALL OTHER INPUTS FLOATING.

THE ABOVE CHARACTERISTICS APPLY ONLY IF THE FOLLOWING CONDITIONS ARE MET

A) 1/2 HOUR WARM UP REQUIRED UNLESS OFFSET SUBSTRACTED IN SOFTWARE.
B) INPUT CONDITIONING CARD IC3600ATCA PRECEEDING AMP CARD.

INPUTS

P+(3), P-(4)	DIFFERENTIAL ANALOG INPUTS. POSITIVE ON P+ RESULTS IN A NUMBER AT A/D CONVERTER OUTPUT.
ADDR4(48) ADDR5(47) ADDR6(46)	STRAP "ZEROS" TO COM, "ONE" IS OPEN
A41(37) A51(36) A61(35)	CMOS LOGIC INPUTS. ADDRESS "ZERO" IS 000. ADDRESS "7" IS 111.
P15A(27)	NON ISOLATED POSITIVE ANALOG POWER SUPPLY BUSS 3 MA
ACOM(2)	NON ISOLATED ANALOG COMMON
N15A(29)	NON ISOLATED NEGATIVE ANALOG POWER SUPPLY BUSS 3 MA
P15AI(20)	ISOLATED POSITIVE ANALOG POWER SUPPLY BUSS 12 MA
ACOMI(19)	ISOLATED ANALOG COMMON
N15AI(21)	ISOLATED NEGATIVE ANALOG POWER SUPPLY BUSS 10 MA

OUTPUTS

CH+(14)	10V ANALOG OUTPUT/2K LOAD 400PF - POSITIVE ON P+(3) IS POSITIVE ON CH+(14).
CH-(15)	ACOM RETURN FOR CH+(14).
AC+(8), AC-(9)	DIFFERENTIAL ANALOG/ 100K LOAD - POSITIVE ON P+(3) IS POSITIVE ON AC+(8).
OPERM (25)	CHANNEL ENABLE CMOS LOGIC SIGNAL.A LOGIC ZERO WILL ENABLE BOTH CARDS THAT ARE IN THE SAME CHANNEL.

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REV 3	MADE BY AZRA EHSAN			I.C. NO.		CONT ON SH. 44 SH. NO. 43

2.2.1-4 IC3600 AEXA EXPANDER CARD

THIS CARD IS USED TO EXPAND A CHANNEL AMPLIFIER TO MORE THAN ONE ROW IN THE ANALOG INPUT RACK. FUNCTIONALLY THE CARD PROVIDES MULTIPLEXING OF ANALOG INPUT SIGNALS IN THE ROW THIS IS MOUNTED, INTO A CHANNEL AMPLIFIER THAT SERVES MORE THAN ONE ROW OF ANALOG INPUTS. THUS THE AUXILLARY CHANNEL SWITCHES ON THIS CARD PROVIDE THE ADDITIONAL LEVEL OF SWITCHING REQUIRED TO ISOLATE DIFFERENT ROWS OF POINT SWITCHES. ADDRESSING OF THE SWITCH IS CONTROLLED BY A COINCIDENCE GATE AND DIODES. EACH OF THE 8 ROWS (MAX.) IN THE ANALOG INPUT STICK HAS A DIFFERENT STRAPPING. THESE STRAPS ARE ADDR4, ADDR5, ADDR6. THE COINCIDENCE OF THE ADDRESS BITS A41 (34), A51(33), A61(32) WITH ADDR4, ADDR5, AND ADDR6 RESPECTIVELY ENABLES THE AUXILLARY SWITCH, AND CONNECTS ONE AND ONLY ONE ROW OF ANALOG POINT SIGNALS TO A CHANNEL AMPLIFIER. ADDITIONAL CIRCUITRY ON THE CARD PROVIDES PROTECTION FOR THE CMOS LOGIC AND THE CMOS SWITCH. THIS CARD CONTAINS ALSO THE NECESSARY JUMPERS FOR POINT ADDRESSES, CHANNEL LINES, AND P15 DCOM TO CONNECT THESE LINES FROM THIS ROW TO ROW ABOVE.

INPUTS A41(34,37), A51(33, 36), A61(32, 35): AUXILLARY CHANNEL ADDRESS LINES, ISOLATED (WIRED ON THE AIR TO THE P.S. CARD, OR JUMPERED TO THE P CONNECTOR). CMOS INPUT.

INPUTS ADDR4(48), ADDR5(47), ADDR(46): STRAP INPUTS FOR ADDRESS COINCIDENCE. (WIRED ON THE AIR). CMOS INPUT.

OUTPUTS P+(3), P-(4): MULTIPLEXED LINE OF ANALOG INPUT SIGNALS. TIED ON THE AIR TO CHANNEL AMPLIFIER INPUT.
ANALOG VOLTAGE - ±12.5V MAX.

ØPERM (25) OUTPUT

CMOS OUTPUT WITH 20K SERIES RESISTOR, ZERO WHEN AUXILLARY CHANNEL SWITCH IS CLOSED. MAY BE USED TO CLOSE POINT SWITCHES.

BUS

+15V	-15V
5MA MAX.	4MA MAX.

REV 1	REV 4	REV. 5	PRINTS TO	APPROVALS Azra Ehsan	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED Jan 25/79			FIRST MADE FOR		ITS APPLICATION NOTES
REV. 3	MADE BY AZRA EHSAN			I.C. NO.		277A4936
						CONT ON SH. 45 SH. NO. 44

2.3.0 OVERTEMPERATURE TRIP MODULE DS3820NOT

THE OVERTEMPERATURE TRIP MODULE, DS3820NOT IS THE PRIMARY OVERTEMPERATURE PROTECTION SYSTEM FOR THE GAS TURBINE IMPROVED TEMPERATURE SYSTEM. THREE EXHAUST TEMPERATURE THERMOCOUPLES ARE MONITORED. THE MEDIAN THERMOCOUPLE SIGNAL IS SELECTED AND COMPARED TO AN ADJUSTABLE REFERENCE (900°F TO 1200°F). IF THE TEMPERATURE IS HIGHER THAN THE REFERENCE, A FAULT SIGNAL IS LATCHED AND SENT TO THE TURBINE CONTROL PANEL INDICATING OVERTEMPERATURE.

THE NOT MODULE CONSISTS OF AN OVERTEMPERATURE TRIP PWB, DS3800NOTA, MOUNTED INSIDE A METAL ENCLOSURE. SINCE ALL OF THE ELECTRONICS FOR THE MODULE ARE ON THE NOTA PWB, ALL OF THE FOLLOWING REFERENCES WILL BE TO THE NOTA PWB AND NOT TO THE MODULE.

2.3.1 DS3800 NOTA OVERTEMPERATURE TRIP CARD

THE NOTA (OVERTEMPERATURE TRIP) IS A SINGLE-SIZE PRINTED WIREBOARD DESIGNED ESPECIALLY FOR THE IMPROVED TEMPERATURE SYSTEM OVERTEMPERATURE TRIP MODULE-DS3820NOT.

THIS CARD MONITORS THREE THERMOCOUPLES AS A PART OF AN OVERTEMPERATURE PROTECTION SYSTEM. THE MEDIAN THERMOCOUPLE SIGNAL IS SELECTED AND COMPARED TO AN ADJUSTABLE REFERENCE. IF THE TEMPERATURE IS HIGHER THAN THE REFERENCE, A FAULT LATCH IS SET TO INDICATE OVERTEMPERATURE. AN EXTERNAL MANUAL RESET IS REQUIRED TO CLEAR THE FAULT LOGIC.

MAJOR CIRCUIT FUNCTIONS ARE:

THERMOCOUPLE CONDITIONING AND FILTERING.

THREE INSTRUMENTATION AMPLIFIERS TO AMPLIFY LOW LEVEL THERMOCOUPLE SIGNALS.

COLD JUNCTION COMPENSATION (INCOMING THERMOCOUPLES ARE TERMINATED ON THIS CARD).

MID-VALUE SELECT CIRCUIT.

LEAD-LAG FILTER NETWORK.

TEMPERATURE STABLE REFERENCE.

ADJUSTABLE GAIN AMPLIFIER TO SET TRIP POINT.

OVERTEMPERATURE COMPARATOR.

FAULT LATCH WITH EXTERNAL RESET.

BUFFERED FAULT LOGIC OUTPUTS AND DRIVER FOR EXTERNAL "OT TRIP" LED.

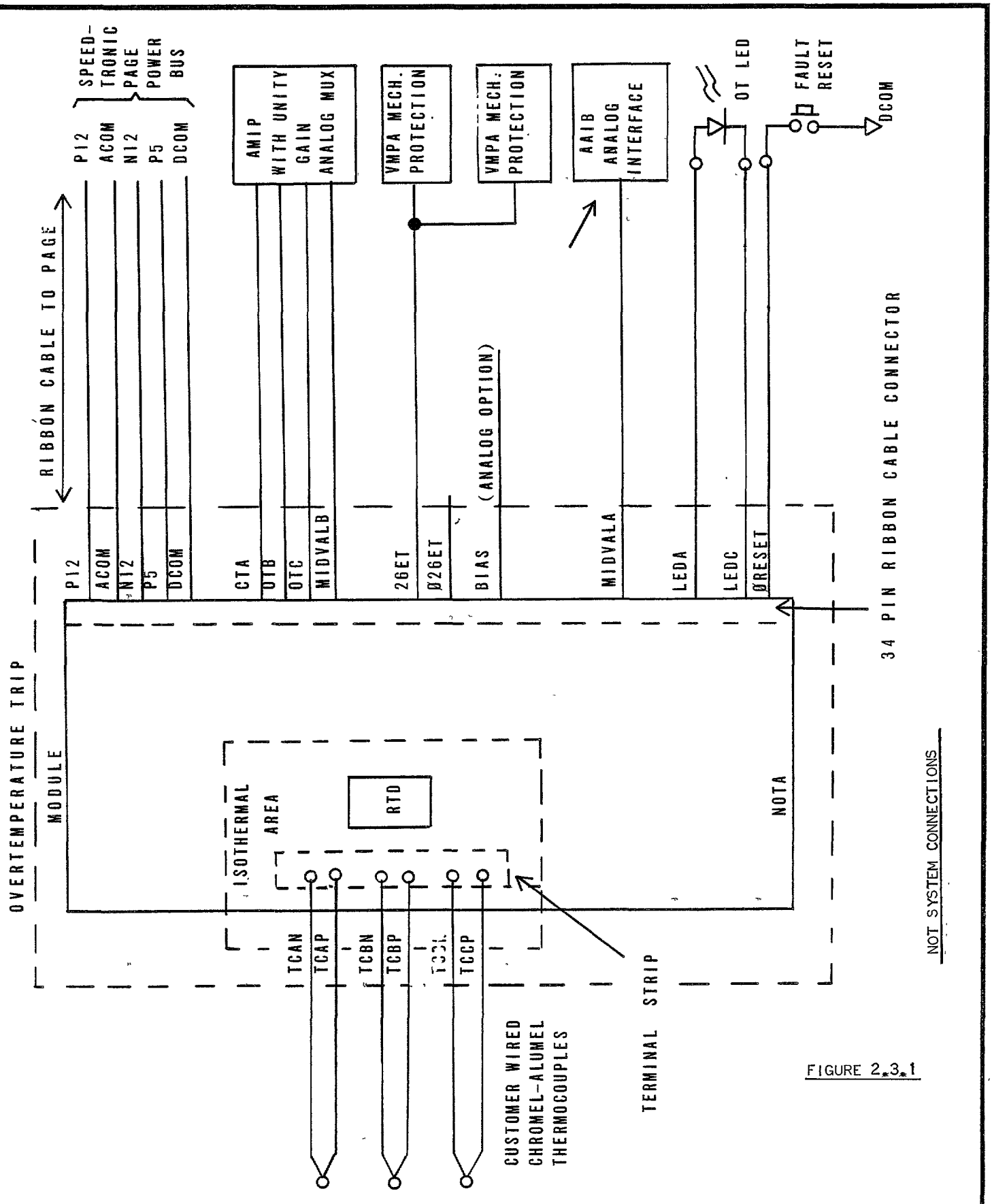


FIGURE 2.3.1

REV. 1	REV. 4	REV. 7	PRINTS TO	ENGINEER Azra Ehsan	GENERAL ELECTRIC	ITS APPLICATION NOTES
REV. 2	REV. 5	ISSUED Jan 25/79			DRIVE SYSTEMS DEPT.	277A4936
REV. 3	REV. 6	MADE BY AZRA EHSAN			SALEM, VA. U.S.A.	CONT. ON SH. 47 SH. NO. 46

DSD 0151 (09-76)

A SINGLE 34 CONDUCTOR RIBBON CABLE CONNECTS NOTA TO THE SPEEDTRONIC* CONTROL PAGE THROUGH DS3600XJBA INTERCONNECT CARD. ALL NOTA POWER IS SUPPLIED FROM THE SPEEDTRONIC* CONTROL PAGE VIA THE RIBBON CABLE. NOTA DOES NOT HAVE AN 80 PIN EDGE CONNECTOR. THREE CUSTOMER CONNECTED #18 GAUGE TYPE K CHROMEL ALUMEL THERMOCOUPLES WILL CONNECT TO A SIX POINT PWB TERMINAL STRIP THAT IS PART OF THE NOTA CARD. THE NOTA CARD CONTAINS NO ELECTOSTATIC SENSITIVE COMPONENTS.

ALL POWER AND OUTPUT SIGNAL CONNECTIONS FROM THE NOT MODULE ARE MADE VIA THE NOTA CONNECTOR PLUG JA (34 PIN RIBBON CABLE CONNECTOR).

P12	JA2
N12	JA4
ACOM	JA1
P5	JA34
DCOM	JA33

THERMOCOUPLE CONNECTIONS

1. WITH THE MODULE COVER REMOVED, WIRE THE THREE OVERTEMPERATURE THERMOCOUPLES TO THE TERMINAL STRIP TS.

THERMOCOUPLE	CONNECTION POINT
OTA	YELLOW TO TCAP (TS2) RED TO TCAN (TS1)
OTB	YELLOW TO TCBP (TS4) RED TO TCBN (TS3)
OTC	YELLOW TO TCCP (TS6) RED TO TCBN (TS5)

2. USE CHROMEL-ALUMEL TYPE-K (#18 GAUGE) THERMOCOUPLES ONLY. 500 FT. MAXIMUM LENGTH.

3. TREAT THERMOCOUPLE WIRING AS LOW LEVEL.

THE OVERTEMPERATURE TRIP MODULE IS NORMALLY FACTORY CALIBRATED AS PART OF THE IMPROVED TEMPERATURE SYSTEM, AND THUS NO START-UP CALIBRATIONS ARE REQUIRED.

*TRADEMARK OF GENERAL ELECTRIC COMPANY, U.S.A

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS Azra Ehsan	GENERAL ELECTRIC	ELEMENTARY DIAGRAM ITS APPLICATION NOTES
REV. 2	ISSUED Jan 25/79			FIRST MADE FOR		277A4936
REV 3	MADE BY AZRA EHSAN			I.C. NO		CONT. ON SH. 48 SH. NO. 47

MAINTAINABILITY

NOTA IS COMPLETELY ENCLOSED INSIDE THE OVERTEMPERATURE TRIP MODULE, DS382ONOT. ACCESS VIA REMOVABLE BUTTONS IS AVAILABLE FOR ADJUSTING THE TRIP POINT SET POTENTIOMETER, R13, AND INSERTING A DIGITAL THERMOMETER INTO THE RTD. SEVERAL MAINTENANCE POINTS ARE AVAILABLE BY REMOVING THE MODULE COVER.

TWO MAINTENANCE POINTS ARE AVAILABLE TO JUMPER OUT THE COLD JUNCTION COMPENSATION (THUS REFERENCING THE CIRCUITRY TO 32°F) TO SIMPLIFY FIELD CALIBRATION. TWO ADDITIONAL MAINTENANCE POINTS ARE AVAILABLE TO SHORT OUT THE LONG LEAD FUNCTION TO PREVENT NUISANCE TRIPS DURING TRIP POINT CALIBRATION. THE SIGNAL MIDVAL IS ALSO AVAILABLE FROM A MAINTENANCE POINT.

FOR BEST ACCURACY, THE TRIP POINT SHOULD BE CHECKED AND RECALIBRATED YEARLY.

REFER TO DS382ONOT OVERTEMPERATURE TRIP MODULE ELEMENTARY FOR THE FOLLOWING:

TROUBLESHOOTING AND COLD JUNCTION CALIBRATION....SECTION 3
MAINTENANCE INSTRUCTIONS.....SECTION 1.6

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS Azra Ehsan	GENERAL ELECTRIC	ELEMENTARY DIAGRAM ITS APPLICATION NOTES
REV. 2	ISSUED Jan 25/79			FIRST MADE FOR		277A4936
REV 3	MADE BY AZRA EHSAN			I.C. NO.		CONT ON SH 49 SH. NO 48

2.4.0 DS3820PS3 POWER SUPPLY

THE POWER SUPPLY CAN OPERATE ON AC OR DC INPUT VOLTAGES. THE DC INPUT VOLTAGE RANGE IS FROM 90 VDC TO 140 VDC, WITH MOMENTARY DIPS TO 70 VDC. THE AC INPUT VOLTAGE RANGE IS FROM 80 VRMS TO 132 VRMS, 47-63 HZ. THE INPUT CURRENT IS 4.7 AMPS RMS MAXIMUM OR 5.5 AMPS DC MAXIMUM (100 A 1 CYCLE INRUSH).

IT PROVIDES PARALLEL OUTPUTS OF:

+5 ± .15 VDC AT 22.5 AMPS, 50MV P.P RIPPLE MAX
+15 ± .15 VDC AT 2 AMPS, 50MV P.P RIPPLE MAX
-15 ± .15 VDC AT 2 AMPS, 50MV P.P RIPPLE MAX
+28 ± .5 VDC AT 1 AMP, 100V P.P RIPPLE MAX

TWO TERMINAL BOARDS TB1 AND TB2 ARE LOCATED ON THE POWER SUPPLY FOR POWER CONNECTIONS AND JUMPERS.

POWER CONNECTIONS ARE MADE AT TB1-1 (HIGH) AND TB1-2 (COMMON).

CHASSIS GROUND IS CONNECTED AT TB1-4.

JUMPERS ON POWER SUPPLY TERMINALS ARE IN THE FOLLOWING CONFIGURATION:

TB1 (4) TO TB1 (5)
TB1 (5) TO TB1 (6)
TB2 (1) TO TB2 (2)
TB2 (7) TO TB2 (8). FOR DC OPERATION, TB2 (7) TO TB2 (6) FOR AC OPERATION.

REGULATED OUTPUT DC VOLTAGES FROM THE POWER SUPPLY ARE AVAILABLE ON TWO PARALLEL OUTPUT CONNECTORS J1 AND J2 LOCATED ON THE RIGHT SIDE OF THE POWER SUPPLY. THE CONNECTOR J1 IS DIFFERENT FROM J2 IN THAT J1 HAS R5H AND R5L REMOTE SENSE LINES USED TO REGULATE THE +5VDC AT THE LOAD. SEM CONNECTOR JA IS CONNECTED TO CONNECTOR J1 OF THE POWER SUPPLY. CONNECTOR J2 MAY BE USED FOR ANY AUXILIARY EQUIPMENT.

THE MONITOR POWER IS ENERGIZED BY THE CIRCUIT BREAKER CB1. THE OUTPUT VOLTAGES ARE LOCATED ON J1 AND J2. OUTPUT CURRENT AND VOLTAGE CAN BE MONITORED VIA TEST POINTS LOCATED ON THE CONTROL CARD (DS3800NPSC). THE POWER SUPPLY HAS OVERVOLTAGE PROTECTION AND SHORT CIRCUIT CURRENT LIMITING ON EACH OUTPUT VOLTAGE. AN MOV PROTECTS THE INPUT AGAINST LINE TRANSIENTS OF SHORT DURATION. A BALUN CIRCUIT CONSISTING OF A SPLIT REACTOR AND TWO CAPACITORS IS USED AS EMI FILTER TO PREVENT SWITCHING SPIKES FROM BEING DEVELOPED ACROSS THE INPUT POWER LINE.

MECHANICAL

1. DIMENSIONS: 111/4"W X 8"H X 81/2"D (SEE OUTLINE DS3820PS3A SH 6ZA)

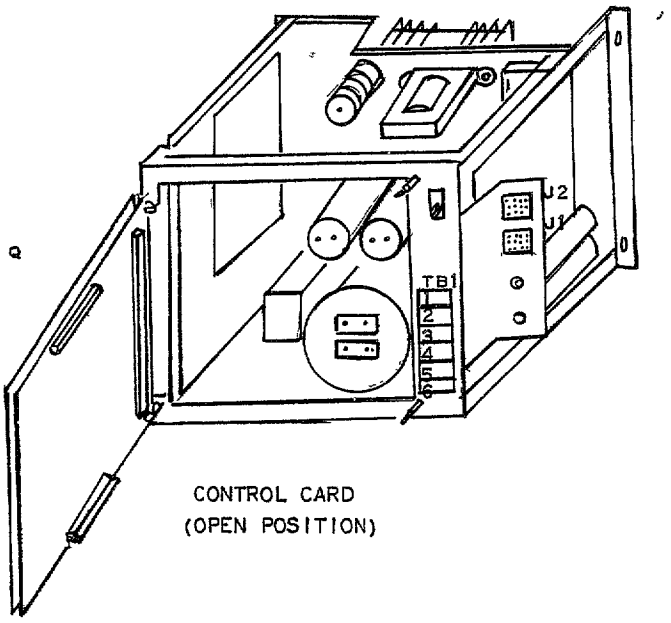
2. WEIGHT: 18 POUNDS

3. MOUNTING: (4) #8 SCREWS, SEE OUTLINE FOR DRILL PLAN.

4. COOLING: CONVECTION

5. CLEARANCES: (MINIMUM)

   A. LEFT SIDE - 2" (AIR FLOW)
   B. TOP -3" (AIR FLOW)
   C. RIGHT SIDE - 3" (WIRING)
   D. BOTTOM-3" (AIR FLOW)
   E. FRONT - 9" (CARD SWING)



CONTROL CARD
(OPEN POSITION)

RECEPTACLES J1 AND J2 MATE WITH PLUG AND PIN 218A4609P1 AND P5

J1	J2
1 - LVSH	LVSH
2 - LVSL	LVSL
3 - R5H	NOT USED
4 - R5L	NOT USED
5 - P5	P5
6 - DCOM	DCOM
7 - DCOM	NOT USED
8 - P28	P28
9 - P5	NOT USED
10- P15	P15
11- N15	N15
12- ACOM	ACOM

*CONNECT TB2-7 TO TB2-8 FOR BATTERY OPERATION OR TB2-6 TO TB2-7 FOR AC INPUTS.

*TB1	
1	DC(P)+ACLH
2	DC(N). ACLL
4	CHASSIS
5	DCOM
6	ACOM

POWER SUPPLY ASSEMBLY
FIGURE 2.4.1

ORDERING
ORDER CATALOG NUMBER AND FORM (DS3820PS3A)

REV. 1	REV. 4	REV. 7	PRINTS TO	ENGINEER Azra Ehsan	GENERAL ELECTRIC	ITS APPLICATION NOTES
REV. 2	REV. 5	ISSUED Jan 25/79				
REV. 3	REV. 6	MADE BY AZRA EHSAN			SALEM, VA. U.S.A.	2 7 7A-4 9 3-6 CONT. ON SH. 51 SH. NO. 50

DSD 0131 (09-76)

REMOTE SENSING LINES ARE AVAILABLE TO ENABLE REGULATION AT A REMOTELY CONNECTED LOAD. SENSOR OUTPUTS ARE UTILIZED TO PROVIDE A LOGIC SIGNAL TO EXTERNALLY CONNECTED ELECTRONICS IF ANYONE OF THE INPUT VOLTAGES FALLS BELOW A SET PERCENTAGE OF NORMAL VOLTS (5% FOR P5, P15 AND N15, 20% FOR P28).

A STATUS LIGHT IS LOCATED ON THE CONTROL CARD AND MONITORS POWER SUPPLY STATUS. WHEN IT IS "ON" ALL POWER SUPPLY OUTPUTS ARE NORMAL.

IF THE POWER SUPPLY IS FUNCTIONING PROPERLY, IT SHOULD ONLY BE NECESSARY TO TRIM THE OUTPUT VOLTAGES AT THE EXTERNAL LOADS. THE FOLLOWING ADJUSTMENTS CAN BE MADE WHILE MONITORING THE LOAD VOLTAGE:

P5 ± .05 VDC	R1 (ON POWER SUPPLY CONTROL CARD)
P15 ± .05 VDC	R9 (ON POWER SUPPLY CONTROL CARD)
N15 ± .05 VDC	R10 (ON POWER SUPPLY CONTROL CARD)
P28 + .5 VDC	R8 (ON POWER SUPPLY CONTROL CARD)

THE POWER SUPPLY WILL DELIVER FULL OUTPUT FOR AMBIENT TEMPERATURES FROM 0 TO 50° C.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS Azra Ehsan	GENERAL ELECTRIC	ELEMENTARY DIAGRAM	
REV. 2	ISSUED Jan 25/79			FIRST MADE FOR		ITS APPLICATION NOTES	
REV. 3	MADE BY AZRA EHSAN			I.C. NO.		277A4936	
						CONT. ON SH. 52	SH. NO. 51

3.0.0 SYSTEMS SOFTWARE PROGRAMS

REFER TO SOFTWARE ELEMENTARY SECTION IN SYSTEMS ELEMENTARY FOR SIGNAL FLOWGRAPH.

3.0.1 THERMOCOUPLE CONDITIONING

THIS PROGRAM SCANS THE THERMOCOUPLES, RTD POINTS AND SHORTED INPUTS (FOR OFFSET COMPENSATION) ON THE ANALOG INPUT RACK. EACH THERMOCOUPLE READING IS COLD JUNCTION COMPENSATED. ANY OFFSET COMPENSATION IS ALSO MADE. THE RESULT IS STORED IN A TEMPERATURE DATA TABLE FOR USE BY OTHER PROGRAMS IN THE SYSTEM CONTROL OR DISPLAY.

3.0.2 COMBUSTION MONITOR

THE COMBUSTION MONITOR IS USED TO DETECT PROBLEMS IN THE COMBUSTION SYSTEM AND EITHER ALARM OR TRIP THE TURBINE IF CERTAIN CONDITIONS OCCUR. THE BASIS OF THE PROGRAM IS TO DETECT A FIRST AND SECOND ORDER SPREAD BETWEEN EXHAUST THERMOCOUPLES AND TO COMPARE THE SPREADS AGAINST A CALCULATED SPREAD LIMIT. THE VARIOUS ALARM AND TRIP POINTS ARE BASED ON THE RATIO BETWEEN ACTUAL SPREADS AND THE CALCULATED ALLOWABLE SPREAD.

3.0.3 WHEELSPACE MONITOR

THE WHEELSPACE MONITOR IS USED TO DETECT HIGH TEMPERATURES IN THE WHEELSPACES OF THE TURBINE. EACH WHEELSPACE HAS A DEFINED UPPER TEMPERATURE LIMIT FOR NORMAL OPERATION. THIS LIMIT IS INCREASED DURING STARTUP AND FOR A PERIOD OF ONE HOUR AFTER COMPLETE SEQUENCE.

EACH WHEELSPACE HAS TWO THERMOCOUPLES. THIS PAIR OF TC'S IS AVERAGED AND THIS AVERAGE IS COMPARED TO THE SPECIFIED LIMIT FOR THAT WHEELSPACE. IN ADDITION, A SPREAD CALCULATION IS MADE BETWEEN THE TWO TC'S IN EACH WHEELSPACE AND COMPARED AGAINST A SPREAD LIMIT. IF ANY WHEELSPACE TEMPERATURE EXCEEDS ITS SPECIFIED LIMIT OR THE SPREAD IN ANY WHEELSPACE EXCEEDS THE SPREAD LIMIT, THE WHEELSPACE ALARM IS SET.

3.0.4 OVERTEMPERATURE ALARM

THE OVERTEMPERATURE ALARM PROGRAM COMPARES THE EXHAUST TEMPERATURE MEDIAN (TXMID) AND THE OVERTEMPERATURE MEDIAN (OTMID) TO CALCULATED ALLOWABLE TEMPERATURES AND GENERATES OVERTEMPERATURE ALARM, TRIP. IN ADDITION AN OVERTEMPERATURE TC SPREAD IS CALCULATED AND COMPARED TO A SPREAD LIMIT FOR AN OVERTEMPERATURE PROTECTION DEGRADED FLAG.

IF TWO OUT OF THE THREE OVERTEMPERATURE THERMOCOUPLES INDICATE BELOW A MINIMUM TEMPERATURE (200°F TYPICALLY) WHILE RUNNING AT RATED SPEED, THEN AN OVERTEMPERATURE SYSTEM FAILURE IS ALARMED.

3.0.5 TEMPERATURE CONTROL

THIS FUNCTION CONSISTS OF TWO PARTS. THE FIRST PART COMPUTES A TEMPERATURE CONTROL REFERENCE FROM A WARM-UP, BASE, OR PEAK REFERENCE AND THE APPROPRIATE VCE AND PCD BIAS. PART ONE ALSO PRODUCES TWO LOGIC OUTPUTS. ONE INDICATES THAT THE REFERENCE IS AT THE WARM-UP VALUE, THE OTHER INDICATES AN ALARM CONDITION IF VCE IS GREATER THAN A FIXED CONSTANT AND VCE BIAS IS LESS THAN PCD BIAS.

PART TWO COMPUTES THE MID VALUE OF THE 13 EXHAUST TEMPERATURE THERMOCOUPLES. THIS VALUE, AND THE SORTED TEMPERATURE ARRAY, ARE MADE AVAILABLE TO OTHER PROGRAMS FOR CONTROL AND DISPLAY. THE REFERENCE DEVELOPED BY PART ONE OF TEMPERATURE CONTROL IS COMPARED TO THE MID VALUE AND THE ERROR SIGNAL IS INTEGRATED TO PRODUCE A TEMPERATURE TRIM RAMP. THE TRIM RAMP IS THEN SUMMED WITH THE PART ONE REFERENCE AGAIN AND OUTPUT AS AN ANALOG SIGNAL. ANALOG HARDWARE SUMS THIS RAMP WITH THE NEGATIVE OF THE MID-VALUE OF THE OVERTEMPERATURE PROTECTION THERMOCOUPLES, PRODUCING AN ERROR SIGNAL WHICH CONTROLS VCE.

THE MID-VALUE TECHNIQUE USED FOR TEMPERATURE FEEDBACK CALCULATIONS PROVIDES AUTOMATIC REJECTION OF THERMOCOUPLES WHICH ARE EITHER TOO HOT OR TOO COLD. THIS TECHNIQUE IS USED FOR BOTH CONTROL THERMOCOUPLES (VIA SOFTWARE) AND PROTECTION THERMOCOUPLES (VIA HARDWARE). THE ERROR TRIM RAMP PROVIDES CORRECTION TO THE MORE ACCURATE MEASURE OF TEMPERATURE (MEDIAN OF 13 TCS) WITH AUTOMATIC FALL BACK TO THE HARDWARE MEASUREMENT (MEDIAN OF 3 TCS) IN THE CASE OF MICROCOMPUTER FAILURE.

3.0.6 INLET GUIDE VANE CONTROL - PART SPEED OPERATION (SURGE)

THE PART SPEED GUIDE VANE CONTROL IS DESIGNED TO CONTROL THE GUIDE VANES DURING STARTUP AND SHUTDOWN OF THE TURBINE TO PREVENT COMPRESSOR SURGE.

THE GUIDE VANE REFERENCE IS A ONE OR TWO SLOPE CURVE WITH UPPER AND LOWER LIMITS. THE SLOPES ARE GENERATED AS A FUNCTION OF HP SET SPEED AND AMBIENT TEMPERATURE. AN ALARM IS GENERATED IF THE ACUTAL GUIDE VANE POSITION DIFFERS FROM THE REFERENCE.

3.0.7 INLET GUIDE VANE CONTROL - PART LOAD OPERATION (TEMP)

THE PART LOAD GUIDE VANE CONTROL IS ACTIVATED ON COMBINED CYCLE JOBS TO MAINTAIN THE EXHAUST TEMP. OF THE TURBINE AT SOME RELATIVELY HIGH VALUE DURING PART LOAD OPERATION OF THE TURBINE. THIS IS DONE TO KEEP HRSG (HEAT RECOVERY STEAM GENERATOR) EFFICIENCY AT AN ACCEPTABLE LEVEL.

IF THIS CONTROL IS ACTIVATED, IT WILL MODULATE THE GUIDE VANES TO A POSITION THAT LIMITS AIR FLOW TO THE PROPER AMOUNT TO MAINTAIN THE EXHAUST TEMPERATURE OF THE TURBINE AT A VALUE DETERMINED BY THE TEMPERATURE CONTROL SYSTEM.

THIS CONTROL ALSO CALCULATES THE COMPRESSOR INLET TEMPERATURE FOR THE PART SPEED GUIDE VANE CONTROL. (NOTE: THIS CALCULATION IS DONE AT ALL TIMES WHETHER THE PART LOAD REFERENCE IS ACTIVATED OR NOT.) AN INLET TEMPERATURE SPREAD IS ALSO CALCULATED FROM THE INLET TC'S AND AN ALARM ACTIVATED IF THIS SPREAD EXCEEDS 25 DEGREES F.

3.0.8 STEAM INJECTION

THE FUNCTION OF THE STEAM INJECTION PROGRAM IS TO CONTROL THE FLOW OF STEAM WHICH IS BEING INJECTED INTO THE TURBINE COMBUSTION CHAMBERS. THE PROGRAM SEQUENCES THE STEAM INJECTION SYSTEM, MONITORS STEAM FLOW, AND REGULATES THE CONTROL VALVE POSITION. THE REFERENCE SIGNAL FOR STEAM FLOW IS EITHER FUEL FLOW, MANUAL CONTROL, OR A START-UP/SHUTDOWN RAMP FUNCTION. STEAM VALVE CONTROL IS ACCOMPLISHED BY CALCULATING REQUIRED DIRECTION OF VALVE TRAVEL AND REQUIRED FREQUENCY OF A PULSING SIGNAL. THE DIRECTIONAL AND PULSE SIGNALS ARE TRANSMITTED TO THE STEPPING MOTOR ACTUATOR ON THE STEAM INJECTION CONTROL VALVE. STEAM PARAMETERS ARE MONITORED, AND ABNORMAL CONDITIONS ARE ALARMED AND/OR GENERATE TURBINE TRIP SIGNALS.

3.0.9 DISPLAY AND PRINTER OUTPUTS

A NUMBER OF SUBROUTINES ARE USED TO DISPLAY AND PRINT (IF PRINTER IS USED) SELECTED DATA POINTS IN ENGINEERING UNITS. THESE POINTS COULD BE DISPLAYED AND PRINTED EITHER SINGULARLY OR IN A PRESELECTED LIST FORMAT. DATA IS INPUT TO THE SYSTEM FROM THE OPERATOR DISPLAY PANEL THUMBWHEEL SWITCHES, PUSHBUTTONS AND KEYSWITCH. FOR DETAILS ON THE LISTS AND POINT DISPLAY REFER TO SECTION 2.1.4.